



International Conference on
Analytics, Articulation & AI: The Prospects for Industry and
Collaboration
(A³I 2026)

March 5th - 7th, 2026



Organized by

Departments of Commerce, English, Mathematics and Computer Science
JAYARANI ARTS AND SCIENCE COLLEGE FOR WOMEN

(Affiliated to Periyar University – Salem)

Nethimedu, Salem - 636002.

Tamilnadu, India

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About Jayarani College:

Jayarani College is a premier educational institution committed to the development and empowerment of young women located in Salem. The college, which is run by the "Immaculate Sisters," was established with the "noble mission" of advancing academic excellence, character development, and value-based education—all of which are embodied in its motto, "Empowering Women." Affiliated with Periyar University, the institution started its academic journey in 2019 and offers seven undergraduate programs in the fields of arts, sciences, and humanities. In 2025, JCS accomplished a noteworthy milestone by securing university clearance to operate an ODL Center within its academic boundaries. By offering postgraduate, diploma, and certificate programs, the ODL Center at JCS equips its student body to handle the challenges of a modern, digitalized world. The academic institution engages in collaborative work with industries through MoUs and has secured MSME Registration under the "Earn While You Learn" program.

About Conference:

Analytics, Articulation & AI: The Prospects for Industry and Collaboration is a premier conference exploring how artificial intelligence and data analytics are reshaping modern business communication. The event convenes industry leaders, AI specialists, data scientists, communication strategists, and business professionals to examine emerging innovations, practical applications, and the evolving landscape of AI-driven communication. Through interactive workshops, panel discussions, and keynote sessions, distinguished experts in AI and communication will share their insights and experiences. Participants will gain actionable knowledge on integrating AI-powered solutions into their communication strategies to enhance organizational effectiveness and commercial outcomes. The conference also provides valuable networking opportunities with peers, executives, and innovators, fostering collaborations that can influence the future of AI in business communication. This event is essential for professionals seeking to remain competitive in an increasingly dynamic digital business environment.

Secretary's Message



"If you dream it, you can achieve it!"

It is my privilege to present the proceedings of the International Conference on Analytics, Articulation & AI: The Prospects for Industry and Collaboration (A³I 2026), held in hybrid mode from March 5th to 7th, 2026.

A³I 2026 serves as an important global platform for researchers, academicians, industry professionals, and innovators to exchange ideas and explore advancements in Analytics, Articulation, and Artificial Intelligence. In an era driven by intelligent technologies and data-centric innovation, fostering collaboration between academia and industry is essential for sustainable growth and societal impact. This conference has successfully created opportunities for meaningful dialogue, knowledge sharing, and strategic partnerships.

The papers included in this volume reflect high-quality research contributions addressing contemporary challenges and emerging opportunities across interdisciplinary domains. Each submission has undergone a rigorous peer-review process to ensure academic integrity, originality, and practical relevance.

I extend my sincere appreciation to the Conference Chair, Organizing Committee, Technical Program Committee, reviewers, keynote speakers, sponsors, and participants for their valuable contributions. My special thanks to the authors whose research forms the foundation of these proceedings.

I trust that this compilation will serve as a significant academic reference and inspire continued innovation and collaboration in the fields of Analytics, Articulation, and AI.

With best wishes for the success of A³I 2026.

Rev. Dr. Jothi Mary FIHM.
Secretary

Principal's Message



“The function of education is to teach one to think intensively and to think critically. Intelligence plus character that is the goal of true education”. -Martin Luther King, Jr.

It is with great pride that I present the proceedings of the International Conference on Analytics, Articulation & AI: The Prospects for Industry and Collaboration (A³I 2026). This conference represents our collective commitment to advancing innovation, research excellence, and meaningful industry collaboration in the era of intelligent technologies.

Analytics and Artificial Intelligence are transforming every sector of society, reshaping how we learn, work, and lead. The strategic integration of data-driven intelligence with effective communication and interdisciplinary collaboration is essential for sustainable progress. A³I 2026 has served as a forward-looking platform that connects academia, industry, and global thought leaders to explore emerging opportunities and address contemporary challenges.

The scholarly contributions featured in these proceedings reflect rigorous research, creative thinking, and practical relevance. They highlight the importance of partnerships that translate knowledge into impactful solutions for industry and society.

I sincerely appreciate the dedicated efforts of the Conference Chair, Organizing Committee, Technical Program Committee, reviewers, keynote speakers, and participants. My special thanks to the authors whose research forms the core of this publication.

May the insights and collaborations initiated through A³I 2026 continue to inspire innovation, responsible AI leadership, and transformative global partnerships.

Dr. Sr. Jenitta Arockiasamy FIHM
Principal

Convener's Message



It is with immense pleasure that I present to you the proceedings of the International Conference on Analytics, Articulation & AI: The Prospects for Industry and Collaboration (A³I 2026), which was conducted in hybrid mode from the 5th to the 7th of March, 2026. A3I 2026 was conceptualized as an evolving platform to create an interface among researchers, academicians, industry experts, and innovators to deliberate on the latest developments in the fields of Analytics, Articulation, and Artificial Intelligence. It is the era of data-driven decision-making and AI-driven intelligence, which is shaping the future of industries and institutions across the globe.

The conference saw enthusiastic and active participation from delegates across the country and the globe, which is a reflection of the global appeal of the theme of the conference. The research articles in this volume offer a range of perspectives, innovative methods, and useful applications to combat today's issues. Each of the research papers was subject to a stringent peer-review process to ensure the quality, originality, and authenticity of the research work. It is our belief that this volume of proceedings will prove to be a useful reference source to the research community, the profession, and the academia.

I would like to express my heartfelt gratitude to the Principal, Conference Chair, advisory committee, technical committee, reviewers, keynote speakers, sponsors, members of the organizing committee, and volunteers who have all contributed to the smooth and successful conduct of the A³I 2026 conference. I would also like to express my special gratitude to the authors of the research papers who have contributed to the scholarly work and participation.

I hope that the collaborative efforts generated through the A³I 2026 conference will continue to promote the ideals of research excellence, innovation, and symbiotic industry collaborations into the future.

Dr. D. Arul Pon Daniel
Convener, A³I 2026

THE EDITORIAL MESSAGE



Dear Readers,

‘When the winds of change blow, some people build walls, and other build windmills.’
- an Ancient Chinese proverb

The editorial team is delighted to put out the proceedings of the International Conference on Analytics, Articulation & AI: The Prospects for Industry and Collaboration (A³I 2026), which was conducted in a hybrid environment from the 5th to the 7th of March, 2026. These proceedings embody the academic contributions of researchers and professionals dedicated to the pursuit of excellence in the field of Analytics, Artificial Intelligence, and the Articulation of interdisciplinary communication. The conference has witnessed a myriad of research contributions to the emerging field of Analytics, Artificial Intelligence, and industry-academia collaboration. Each of the papers published in the conference proceedings has gone through a rigorous peer-review process to ensure the quality of the academic content, its originality, clarity, and its relevance to the challenges of the contemporary world.

We would like to sincerely acknowledge the significant contribution provided by the Conference Chair, the organizing committees, the reviewers, and the authors, whose prompt collaboration has enabled the timely publication of this work. This publication is the result of the combined efforts of all the individuals involved, highlighting the importance of both theoretical and practical contributions. We hope that this publication will prove useful for researchers, practitioners, and students, as well as inspire further research, creativity, and collaboration in the fields of Analytics and Artificial Intelligence.

**Ms. C. Priyadharshini &
Ms. M. Syedalifathima**
Editors - A³I 2026

PREFACE

International Conference on Analytics, Articulation & AI: The Prospects for Industry and Collaboration (A³I 2026) was held from March 5th to 7th, 2026, in hybrid format. The primary purpose of the conference was to facilitate effective communication between academia, industry, and research communities. The conference was a global forum to discuss the latest developments in Analytics, Artificial Intelligence, and communication approaches that promote innovation and sustainable growth.

A³I 2026 was a confluence of researchers, academicians, industry experts, and practitioners who shared their knowledge, presented their original work, and analyzed the latest trends that shape the future of intelligent systems and collaborative ecosystems. The variety of themes covered during the conference highlights the recognition of the need to bridge the gap between theoretical research and practical applications.

The papers presented in these proceedings have been peer-reviewed, showcasing originality, technical content, and practical relevance. We hope that these proceedings will be a valuable academic resource, promoting further research, innovation, and strategic industry engagement in the dynamic world of Analytics and AI.

Thank you for accompanying us on this remarkable intellectual journey.

**Faculty Members & Students,
Departments of Commerce, English Mathematics and Computer Science
Jayarani Arts and Science College for Women, Salem**

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ABOUT JAYARANI COLLEGE

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seven undergraduate programs in the fields of arts, sciences, and humanities.

In 2025, JCS accomplished a noteworthy milestone by securing university clearance to operate an ODL Center within its academic boundaries.

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International Conference on Analytics, Articulation & AI: The Prospects for Industry and Collaboration

Optimizing Real-Time Data Streams for Autonomous Vehicles with Big Data Tools, Swarm-Based Techniques, and Attention Driven Graph Neural Networks

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Abstract: Autonomous vehicles (AVs) rely on real-time processing of massive data streams generated by an array of sensors, including cameras, LiDAR, radar, and GPS. These data streams must be processed with high speed and accuracy to enable AVs to make timely and safe decisions. In this paper, we introducing an integrated framework solution to improve the efficiency and effectiveness of the data stream processing in AVs through using big data tools including Apache Kafka and Spark Streaming with the swarm intelligence-based optimization algorithms, PSO and ACO. The framework aims to address key challenges in real-time data handling, including low-latency processing, accurate decision-making, and dynamic adaptability to varying traffic conditions. The proposed framework, demonstrated a significant improvement in system performance by increasing 30% in processing speed, making the model highly efficient for real-time AV applications. Processing delay was reduced to 40-60 ms compared to over 100 ms in the base system, while end-to-end delay was kept stable between 55-75 ms. These enhancements contribute to a substantial reduction in high-risk prediction errors and an increase in decision-making accuracy, underscoring the framework's potential for improving the safety and reliability of AV systems.

Key words: Autonomous Vehicles, Data Stream Processing, Big Data Tools, Swarm Intelligence, Real-Time Decision-Making.

1 Introduction

As social business sectors have grown and living standards have risen, so too has the number of people who own automobiles[1]. The rise in road traffic brought on modernity has created several transportation and traffic control issues, including severe traffic jams and collisions.

Over the past ten years, a lot of effort has been made to create traffic surveillance systems with the intention of boosting security through on-road scene monitoring. One of the biggest obstacles to the development of autonomous vehicles and smart cities is road safety. Finding a dependable method for predicting accidents before they happen is the goal of accident avoidance, one of the trendiest topics in road safety. Technology for autonomous driving is being developed to lower traffic accidents by substituting an autopilot for human drivers. In order to monitor its surroundings and provide digital images data for the car to make decisions, the self-driving car system uses object recognition[2]. Autopilot analyses data from the multifunctional sensors installed on the self-driving cars to understand the road conditions ahead and respond accordingly [3]. It

is essential to find practical ways to lessen traffic jams in cities, such as anticipating traffic and measuring the whole length of a car's line in the flow of traffic [4]. There are several object detection techniques based on deep learning and machine learning have been used in many different object identification applications, although small item detection suffers from limited resolution. It is imperative to develop a tiny, dependable object identification system that can precisely identify small objects[5].

As the technology for connected autonomous vehicles (CAVs) approaches, researchers and engineers must comprehend the advantages of this emerging field while also planning for the possible traffic disruptions that these cars may bring in the future[6]. Advanced sensor data is possibly available for automated driving system (ADS) cars outfitted with multifaceted sensors (e.g., LiDAR, camera, RTK-corrected global navigation satellite system (GNSS)) for vision[8]. To comprehend the performance of new mixed traffic, a vast variety of data may be retrieved, including ADS operating safety, CAV-traffic interaction, transportation oscillations, heterogeneity, car-following behavior, and lane change behavior[7]. Self-driving cars have countless of data from its numerous sensors and sub-systems every day this calls for efficient methods of processing the real-time processing systems that can effectively analyze the incoming stream of data to guarantee safe vehicle operation[10]. Processing data streams for AVs involves the continuous and real-time handling of vast amounts of data generated by various sensors, such as cameras, LiDAR, radar, and GPS systems to provide the overall picture of the vehicle's surroundings[11]. While speed of decision is important, it is also important that they get it right; and any form of weakness or even lapse may result in poor driving behavior hence danger of an accident. Such requirements cannot be handled effectively with traditional data processing techniques where data is mostly processed in batches or in simple real time modes, resulting in delayed decision-making column or wrong decisions being made.

There are a plethora of uses for machine and deep learning-based object detection in business and academia, such as face identification, pedestrian recognition, logo comprehension, vehicle detection, and many more[3]. On the one hand, it is challenging for autonomous driving cars to handle such a large amount of streaming data given their limited on-board processing capability. On the other hand, latency-sensitive applications like autonomous driving are severely impacted by the significant time delay caused by utilizing a faraway cloud computing facility[12]. Graph neural networks (GNN) can be used to analyze data from the graph structures they take and are recommended for tasks with complicated relational information such as traffic networks in autonomous transportation. GNNs rely on the connections of the graph structure which allows them to extract relations and dependencies between different nodes which in this case are traffic agents such as cars and people. This capacity to capture relations within a spatial-temporal framework is very crucial for motion computing since it helps the autonomous system to predict how several agents within an environment or scenario will move and interrelate. The followings are the proposed research's primary contributions

The study introduces a novel framework that combines Graph Neural Networks (GNN) with attention mechanisms, improving the accuracy and robustness of motion prediction for autonomous vehicles. This research highlights how big data processing frameworks can be integrated with swarm intelligence to fulfil the needs of next-gen self-driven automobiles.

The research demonstrates the effectiveness of applying Particle Swarm

Optimization (PSO) and Ant Colony Optimization (ACO) to enhance model performance, leading to faster convergence and reduced prediction errors.

The study provides substantial evidence that the optimized model reduces high-risk prediction errors and increases decision-making accuracy, contributing to safer and more reliable autonomous vehicle operations.

The following components of the study are required to be completed: Section 2 offers a thorough summary of previous studies. Section 4 looks at the suggested research methodology, while Section 3 provides a comprehensive analysis of the problem description. The results and a thorough analysis are given in Section 5. Section 6 contains the Conclusion and next works. In the context of navigating the AV, attention mechanisms allow the system to focus on crucial traffic agents and important features of the environment hence give higher ranks to such factors when making a decision. By incorporating attention mechanisms into GNNs, the model can better capture the traffic dynamics, and hence, the motion prediction is refined to reduce the risk involved autonomous vehicle navigation.

2 Related Works

The study on an architecture AIBSNF for the purpose of strategically using wearable biosensors, real-time location systems (RTLS), and body sensor networks (BSN) to gather multimodal, high-fidelity, low-noise data [13]. The paper discuss about wearable sensor fusion technologies, RTLS, and uses cases for AI/ML algorithms.

The intelligent defect detection of an ATV tool using signals from several connected sensors is the main topic of this study[14]. A unique technique to data fusion is presented for the detection and diagnosis of operational problems in an ATV. It involves the use of the short time Fourier transform in conjunction with CNN.

The automated ML (AutoML) for domain-specific risk prediction and behavior evaluation presented in this work may be used to autonomous vehicles' (AVs') motion trajectory planning and behavioral decision-making[15]. Using a predictive capacity of 91.7% overall accuracy for four risk categories and almost 95% accuracy for safe-risk differentiation, behavior-based risk prediction using autoML yields good results.

A new, automated process for creating sizable virtual datasets for training and testing is put into practice, coupled with an examination of a machine learning-based radar perception method for object recognition[16]. The virtual dataset is created via electromagnetic modeling based on physics of a complicated scattering environment. Using one channel 77 GHz FMCW radar system, objects are categorized and located within Doppler-Range findings.

Vehicles still employ a CAN bus system to facilitate communication between the many electronic control units (ECUs). The CAN bus, although extremely effective for sending messages, has serious security flaws since it is an unsegmented, unencrypted network without authentication. A comparative study in[17]is carried out across several feature sets using supervised ML algorithms. The CAN intrusion dataset's assault messages are better detected by the RF classifier—which achieves a 97% accuracy rate than by KNN or XG boost.

The research offered a DL-based traffic safety solutions in an ITS with 5G enabled that combines manual and autonomous cars[18]. Experimental results show that the approach successfully addresses the lane change problem in mixed traffic while simultaneously improving accuracy and real-time intention detection, with intention detection rates of 91.58% and 90.88% for left and right lane shifts, respectively.

Problem Statement

The evolution in AV technology has thus witnessed an explosion in the amount of data provided by its sensors and systems. However, with conventional computing, the ability to process information at real-time is a challenge in a way that disrupts the AVs capability in decision making and hence, safety and efficiency[19]. This research aims at addressing this important problem through proposing an integrated framework with enhanced machine learning models to improve the prediction performance and support reliable decisions in autonomous driving scenarios.

3 Autonomous Vehicle Motion Prediction

The approach of this paper uses Lyft Motion Prediction dataset including high-resolution data of traffic agents such as cars, bicycles and pedestrians, is used as data and then pre-process them. This dataset is used as the reference to build up and later test predictive models. Subsequently, a real time data processing framework using Big data tools such as Apache Kafka and Spark Streaming is adopted to deal with the large volumes and high velocity of data streams. The essence of the proposed methodology considers the application of Graph Neural Networks (GNNs) supplemented with an Attention mechanism to analyze traffic agents' interactions and accurately foresee their behavior in the near future. The use of these techniques is to promote the improvement of the decision making of the autonomous vehicle in real-world realistic traffic scenarios. It is illustrated in below Figure 1.

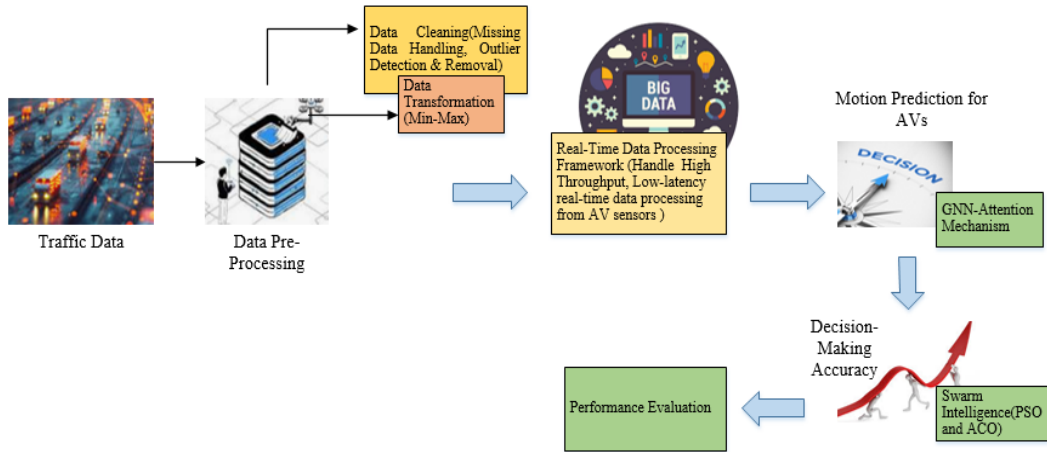


Figure 1: Methodological Diagram of the Proposed Method

3.1 Data Collection

The Lyft Motion Prediction dataset in CSV structure constitutes a complete and detailed record of traffic agent motion required to calibrate sophisticated motion predictions for self-driving cars[20]. This includes what is often referred to as timestamp, whereby the record holds or acknowledges the occurrence of an event at a specific time, thus ensuring that the record can produce a temporal analysis of the movements of the agents. Its main parameters including the position and velocity of each agent at different time intervals is used to determine their subsequent movements and the nature of their interplay.

The nature of traffic scenes and the large number of frames make it possible to conduct a detailed analysis of the efficiency of the data processing and decision-making

algorithms in a real traffic environment, which will ultimately allow enhancing the reliability of self-driving car technologies. Table 1 presents a structured overview of the key features of the dataset, emphasizing their relevance to motion prediction and autonomous vehicle systems.

Table 1: Sample Data from the Dataset

Component	Description	Importance
Timestamp	Records the exact time an event occurs.	Enables accurate temporal analysis of agent movements.
Traffic Agents	Includes various agents such as cars, cyclists, and pedestrians.	Supports the development of robust models by providing data on diverse traffic scenarios and interactions.
Position	Records the position of each traffic agent at different time intervals.	Crucial for predicting future movements and understanding spatial dynamics.
Velocity	Captures the speed of traffic agents at various time points.	Essential for analysing and forecasting the velocity and trajectory of agents.
Contextual Information	May include data on road conditions, environmental factors, etc.	Provides a comprehensive view of the driving environment and enhances predictive accuracy.
High Frequency Data	Dataset captures data at high intervals to simulate real-time scenarios.	Facilitates real-time processing, analysis, and decision-making systems for AVs.

3.2 Data Pre-Processing

4.2.1 Data Cleaning

Missing Data Handling: Another common problem of datasets used in autonomous vehicles is a presence of missing data points caused by sensor failures or problems with the transmitted signal. In Lyft Motion Prediction dataset, the missing values were then imputed using statistical techniques as mean or median in case of numerical values and modes for categorical variables to keep the data as much accurate as possible.

Outlier Detection and Removal: Outliers, or data points that significantly differ from the rest, were detected using methods such as z-score analysis and interquartile range (IQR). These outliers were then removed or treated to prevent them from skewing the analysis.

4.2.2 Data Transformation

Normalization techniques ensures that the dataset's lowest and highest values are changed to 0 and 1, respectively[20], and that all other data points are rescaled linearly to the range of [0, 1] by using Min-Max normalization to guarantee scale uniformity across various features (such location coordinates and velocity)[21].By ascertaining the minimum and maximum values of the feature or column, eliminating the lowest value, and dividing by the range of values, the normalization procedure adjusts each data point independently.[22]. Min-max normalization is represented by Eqn. (1).

$$Z_{norm} = \frac{Z - Z_{min}}{Z_{max} - Z_{min}} \quad (1)$$

Since this normalization technique maintains consistency across the features and enhances the machine learning model's performance during training, it is especially helpful when the features have varying scales.

4.3Real-Time Data Processing Framework Using Big Data Tools

To enhance the swarm intelligence algorithm's performance in interpreting the overwhelming quantities of real-time data that autonomous vehicles produce, this study employs Big Data technologies, including Apache Kafka and Spark Streaming. Big Data tools become necessary for processing the big volume, speed and heterogeneity of data flows generated by LiDAR, cameras and GPS.

Processing the same amount of data as the prior architecture in micro-batches and taking much less time between data arrivals, Spark Streaming ensures the vehicle reacts promptly to changing conditions in the environment. For instance, the proposed framework progresses route planning optimization algorithms which alter the route decisions caused by real-time traffic information so that the vehicle can make the right decision, safely and efficiently.

4.4 Model Selection: Graph Neural Networks (GNNs) with Attention Mechanism

Graph Neural Networks (GNNs)

Given the complexity of traffic environments, where the interactions between various agents (vehicles, cyclists, pedestrians) are essential for accurate motion prediction, Graph Neural Networks (GNNs) are particularly suited for this task. GNNs can model the relational dependencies among these agents by representing them as nodes in a graph, with edges capturing their interactions. An Attention mechanism is used to focus on the most relevant features and to allow the model to dynamically weigh the importance of different nodes and edges in the graph, thereby improving the prediction of agent trajectories.

The architecture of the GNN with an Attention mechanism is as follows:

Input Layer: The input consists of a graph $G = (V, E)$. Each node v_i has features such as position, velocity, and contextual information (e.g., road conditions).

V represents the set of agents (nodes) in the traffic environment. Each agent could be a vehicle, cyclist, pedestrian, or any other relevant entity in the scenario.

E represents the interactions (edges) between these agents, such as proximity or trajectory overlap.

The graph learning layer's process was created with the purpose of creating an undirected graph and connections. This convolution operation combines a feature extracted from a node's neighbors with the importance of these neighbors and modifies the representation of the node. It allows for capturing local dependencies and relational information in a traffic environment occurring at a particular point in time. After that the model starts the information passing through the constructed graph with the help of several consecutive Graph Convolutional Layers (GCLs). Layered on top of the node are the layers that combine information received from a node's neighbors to calculate the new representation for each node to learn the effect of the surrounding agents on the state of the current agent.

Algorithm 1: Training the Proposed GNN with Attention

Input: Given dataset X , Adjacency matrix A , Learning rate η , Number of iterations t , Attention mechanism parameters α (weights), Optimization algorithms: Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO)

Output: Predicted Output Matrix Z and optimized graph structure

1. **Initialize** Weights W^0, W^1, b^0, b^1 , Attention weights α

2. **Initialize** Swarm Optimization (PSO):Set PSO parameters: number of particles, inertia weight, cognitive and social coefficients

3. **Initialize** Ant Colony Optimization (ACO):Set ACO parameters: number of ants, evaporation rate, pheromone deposit

for iteration = 1 to t:

4. **% Forward Propagation**

Apply Attention Mechanism: $A_{att} = \text{Attention}(A, \alpha)$

Layer2_output = $X \cdot A_{att} W^0 + b^0$,

Z = Layer3_output = $\text{Layer2_Output} \cdot A_{att} \cdot W^{(1)} + b^{(1)}$

5. Loss Calculation, $\text{loss} = \frac{1}{2} (X^1 - Z)^2$

6. **% Back Propagation**

7. Optimization Using PSO:

Calculate the fitness of each particle based on the current loss.

Update particle positions and velocities using PSO:

Apply the best parameters found by PSO to the GNN model.

8. Graph Structure Optimization Using ACO:

Update the pheromone levels on the edges based on model accuracy and loss.

Modify the graph structure by reinforcing or weakening edges according to updated pheromone levels:

9. Iteration End:

Continue iterating until convergence or until reaching the maximum number of iterations t

Return Final Outputs

end

return Z

Return the optimized graph structure incorporating changes made through PSO and ACO.

GNN works over several layers where each layer updates the node representations by gathering the information from the neighboring nodes is mentioned in Eqn. (2)

$$h_i^{(l+1)} = \sigma \left\{ \sum_{j \in N(i)} \frac{1}{\sqrt{d_i d_j}} W^{(l)} h_j^{(l)} + b^{(l)} \right\} \quad (2)$$

Where σ is function of activation, $N(i)$ is set of neighbour nodes, W^l is weight matrix b^l is bias vector and $h_i^{(l+1)}$ is feature vector. This convolution operation aggregates the features from a node's neighbors, weighted by their importance, and updates the node's representation.

Attention Mechanism

To enhance the GNN's ability to focus on the most critical interactions in the graph, we incorporate an Attention mechanism. The Attention mechanism calculates a weight a_{ij} for each edge between nodes i and j indicating the importance of the

interaction between these nodes.

The attention score e_{ij} for an edge is computed using Eqn. (3)

$$e_{ij} = \text{LeakyReLU}(a^T [Wh_i || Wh_j]) \quad (3)$$

Where, W is shared weight matrix applied to the feature vectors of nodes i and j and a is a learnable attention vector that determines how much attention should be paid to the connection between nodes. Softmax function used to produce the final attention weights by normalization.

During the decoding process, distinct attention is given to every segment of the input sequence. More precisely, in the attention mechanism, the activation from a bi-directional unit is represented by α' . When predicting the output x_t , the amount of attention x_t should pay to a' from the encoder network is represented by $\alpha_{tt'}$, which is calculated as follows in Eqn. (4)

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in N(i)} \exp(e_{ik})} \quad (4)$$

With the attention weights computed, the model proceeds to update the node representations using Graph Attention Layers (GALs). In these layers, each node aggregates the features of its neighbors, weighted by the attention scores.

This weighted aggregation helps the model to put more weight on relevant neighbors which is very important in traffic environment because some interactions such as pedestrian nearby may be more important than vehicle far away. The enhancement of the agents' dependencies and interactions are improved through the design of the graph attention layers, a Readout operation is applied to the node features after several graph attention layers, which makes the predicted values more precise and subjects them to the actual context.

The graph-level representation at the aggregation level concerns the state of the traffic environment inclusive of all the agents as well as interactions. This representation is then passed through one or more fully connected (dense) layers to map it into the output space which is required.

GNN-Attention architecture is the output layer needed to predict the future behaviors of the agents in the traffic scene. These are the predictions, which characterize direction and velocity of each agent, and are essential for decision-making systems of AV's. By correctly predicting nearby agents' behavior, the vehicle can assess possible interactions and prepare actions to avoid conflicts. Further, the scalable architecture of the model takes into consideration, contextual cues of the environment including the state of the road and traffic signals which are then used to make a better estimate of agent tendencies to provide the accurate and reliable motion prediction in the autonomous driving system.

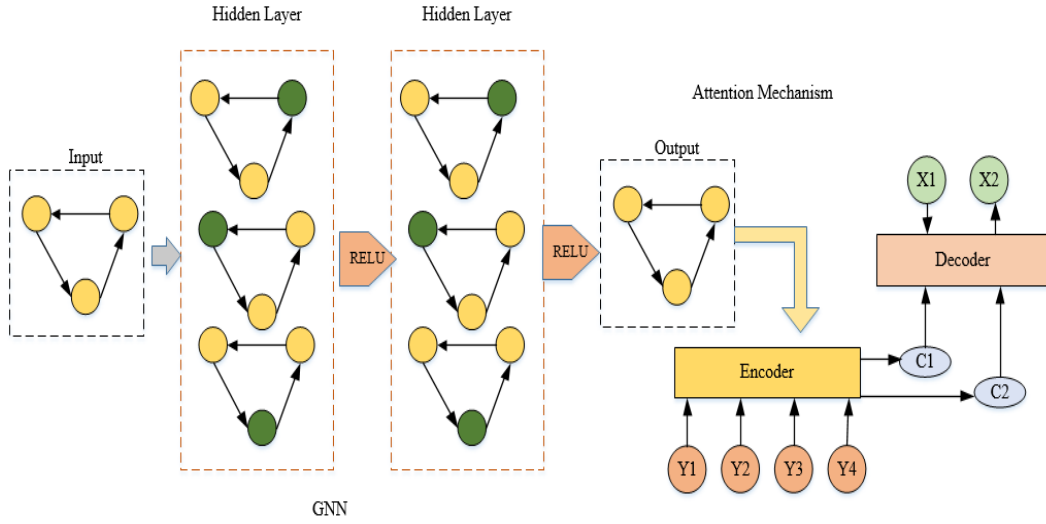


Figure 2: GNN-Attention Mechanism Architectural Diagram

4.5 Swarm Optimization Techniques for Enhancing Model Performance

Once the model architecture is established, swarm-based optimization techniques are applied to fine-tune the model's parameters. Two popular swarm optimization algorithms are considered:

Particle Swarm Optimization (PSO): The present work relates to an optimization algorithm known as PSO which is a population-based algorithm miming a bird flocking procedure. In PSO, each potential solution is called a “particle” within the search space, and its position as well as velocity is updated according to the experience of the particle and of close neighboring particles. Thus, using PSO’s optimization potential, the tracking algorithm may optimize the coordinates of the predicted item location taking into account other factors like motion models, past data and real-time measurements. The recurrent optimization process of PSO refines the predicted positions to take into consideration errors and uncertainties that arise from object classification and initial tracking.

Particle Swarm Optimization (PSO) seeks the optimal value for each round by having particles fly randomly across the screen. The term p_{best} refers to the particle's fittest value throughout each cycle of search. The optimal solution, or fitness value, is calculated and saved after the fitness function is applied to each and every particle. PSO, which has a location known as l_{best} , maximizes the best population value obtained relative to any particle in the neighborhood.

A sensitive particle measures each of the created populations as topological neighbors. The best value from the executed population is selected, and that sensitive best value is identified as the optimal solution as g_{best} . Every time a particle is released, the PSO tries to adjust its speed such that it approaches p_{best} and l_{best} . The speed in the direction of the p_{best} and l_{best} locations is defined by arbitrary words that have arbitrary produced amounts.

The optimization of the GNN-Attention model's parameters is crucial for enhancing its predictive accuracy and computational efficiency. PSO controls the timing parameters of batches according to the arrival rates of real time sensor data. Thus, through the use of PSO, the system can scale up or down the data rates and improve on the latency and processing capabilities of the system. The real time adjustment option improves the entire functioning of the data processing system since it enables efficient

response to varying data rates for optimum system stability and throughput. This process will try to reduce the loss function of the model hence common being Mean Squared Error (MSE) that is the dissimilarity between the predicted and the true agent trajectory. The MSE is computed as using Eqn. (6)

$$MSE = \frac{1}{n} \sum_{i=1}^n ((x_i^{Pred} - x_i^{true})^2 + (y_i^{Pred} - y_i^{true})^2) \quad (6)$$

Ant Colony Optimization (ACO): ACO is yet another swarm-based technique derived from the foraging behavior of ants. In the case of motion prediction, ACO can be applied for optimization of the sequence of the predicted positions SOPT to minimize the total error in the trajectory.

Finally, ACO is also used to improve GNN-Attention model from the structural level involved in the graph optimization level. Building on the principle used by ants in a colony to search for the best possible food source, ACO isolates the most significant edges in the graph that denotes the relationship between nodes or agents. In nature, ants put their scent, the pheromones on the paths that lead to the foods which are accessible for them and put more pheromones more often on the shorter paths so that the paths will attract more ants. Likewise, considering the GNN, ACO mimics this action by cyclically altering the probability value of the edges according to the model's accuracy. The connections that contribute to better forecast are strengthened and boost during each epoch while the unimportant links faded or removed those leads to a simplified graph structure.

The training part of the GNN-Attention model makes the right predictions based on the input information by feeding pre-processed data including the traffic environment and the state of many agents into the model. This data is then fed to the GNN-Attention architecture which can capture the graph attention layers and is able to dynamically weigh the neighboring nodes for interactional importance. The weights of the presented neural network and the structure of the graph are adjusted during training in order to minimize an entropy loss function defined for predicting the agent's trajectory with the Mean Squared Error (MSE) as the target loss. To this end, methods of swarm intelligence, for example PSO and ACO are incorporated into the training phase. PSO avoids training phase and optimizes the model's parameters including learning rates, number of attention heads, and depth of the network, to exhaust all the best combination of values. ACO, on the other hand, continuously adjusts the graph's structure, identifying which edges should be emphasized or discarded based on their contribution to the model's accuracy.

4 Results and Discussion

In this section, the study presents and analyzes the results of our experiments using the Lyft Motion Prediction dataset to evaluate the performance of the proposed AV motion prediction framework. The analysis focuses on key performance metrics, including prediction accuracy, processing speed, and the GNN-Attention model. The simulation environment was built using Python programming environment based on the Windows 10 system which provided realistic traffic dynamics and supported big data tools integration.

5.1 GNN-Based Motion Prediction Layer Verification

The offline verification method is designed to verify the effectiveness of the GNN with attention mechanism on real-time motion prediction for AVs.

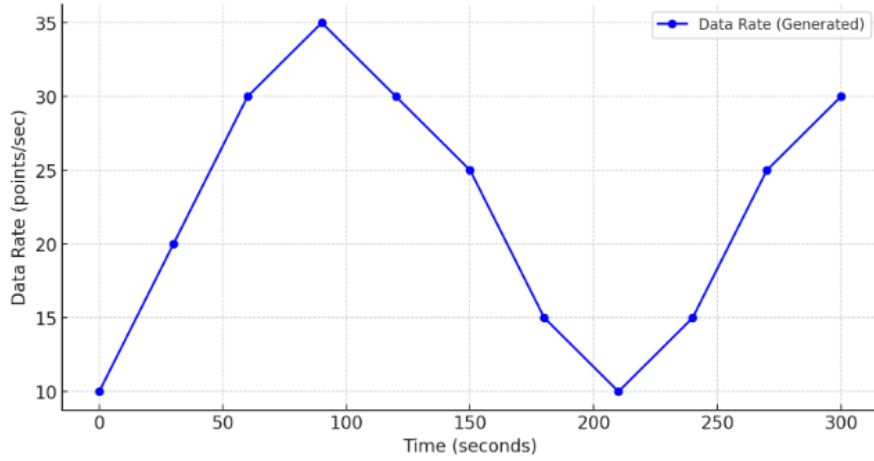


Figure 3: Data Rate Generated by the Data Generator

Figure 3 shows the data rate generated over time, monitor the data rates every **30 seconds** using a **PSO-optimized traffic monitor**, and the real-time data rates are processed using the GNN model. Figure 4 shows the relative error percentage between y_{origin} and $y_{predict}$. The average relative error for the day is approximately 2.5%. The maximum relative error appears around the 18th sampling point and is 12%.

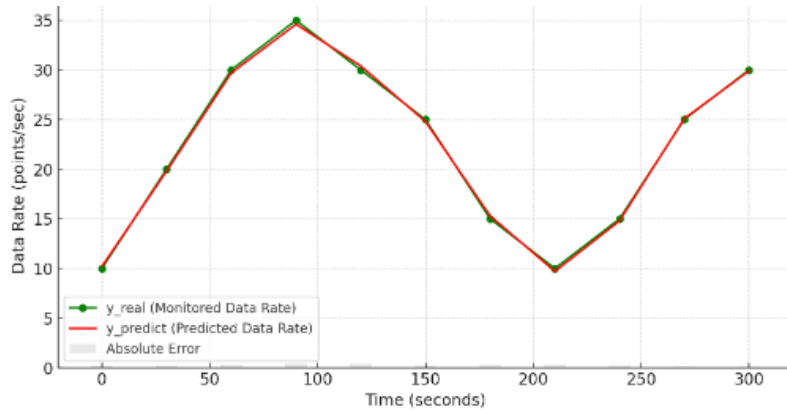


Figure 4: Monitored Data Rate- Fitting Result (GNN+PSO)

Figure 4 shows the real data rate observed (y_{real}), the predicted rate ($y_{predict}$), and the absolute error (abs_error). The **average prediction error** using GNN is approximately **1.8%**. The **maximum error** recorded is **4.5%**.

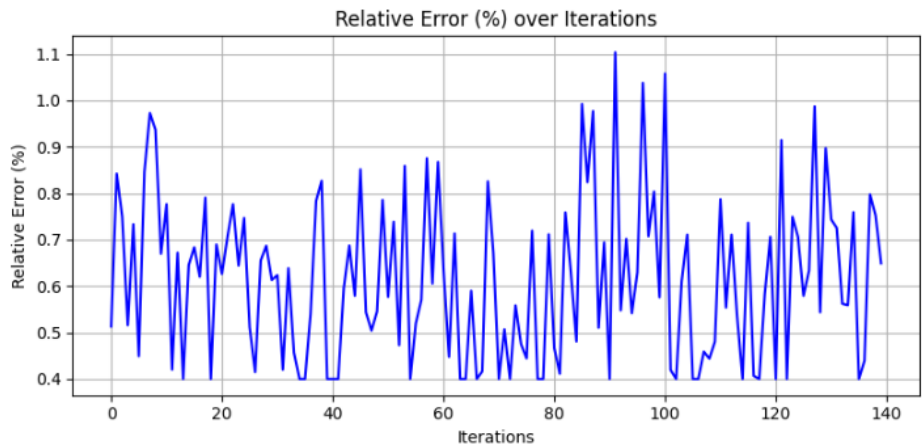


Figure 5: Relative Error

Figure 5 displays relative error over iterations, showing fluctuations capped at 1.33% and a minimum threshold of 0.4%. The average relative error is approximately 0.63%, indicating moderate error levels with some consistency. This visualization is useful in determining the stability and accuracy of errors as well as the performance of each iteration.

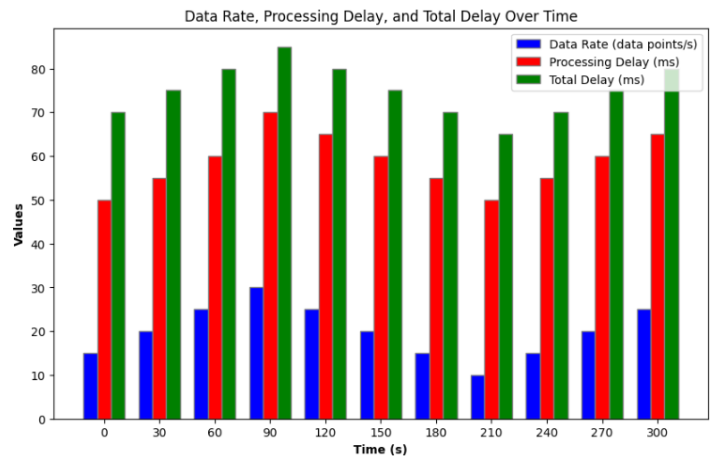


Figure 6: Data Rate, Processing Delay and Total Delay

The Figure 6 compares data rate, processing delay, and total delay over time. It shows distinct patterns where the data rate fluctuates between 10 and 30 data points/s, while both processing and total delay exhibit a steady rise and fall, peaking around 70-85 ms. This visualization offers insight into the system's performance and responsiveness to varying data rates.

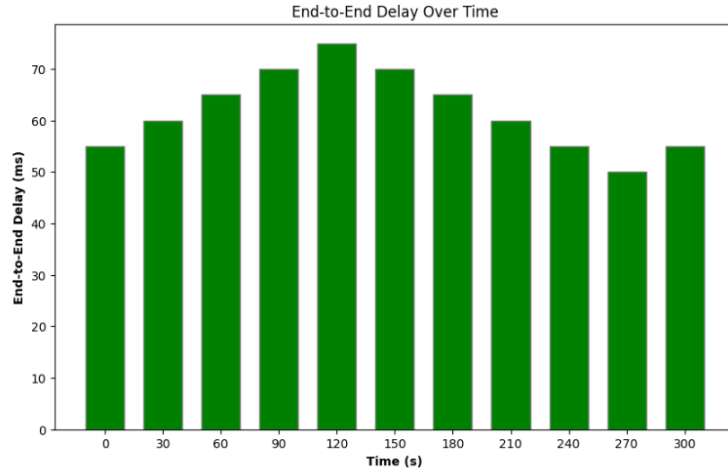


Figure 7: End-End Delay over Time

In Figure 7 fluctuations in processing delay over time, with values ranging from approximately 3250 ms to 4750 ms. The graph depicts the oscillatory pattern contaminated with noise which suggests that the delay varies periodically. This visualization is useful in comprehending spatial elasticity and temporal constancy of the processing delay in the system.

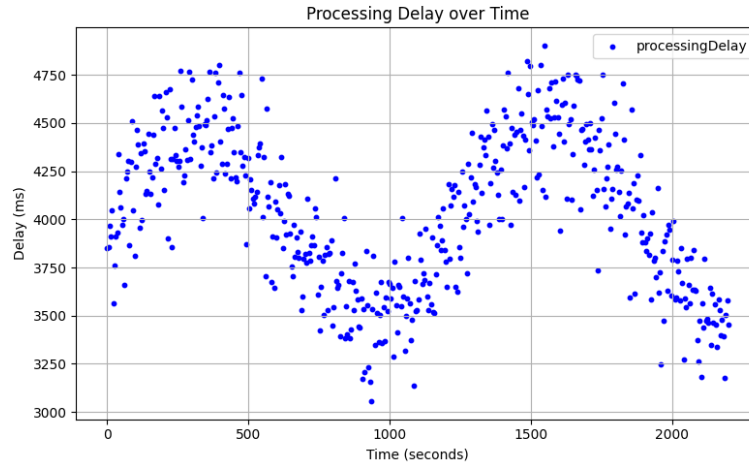


Figure 8: Processing delay with the optimized framework

Figure 8 illustrates that the system workload remains stable around 0.90, demonstrating effective data processing management with minimal fluctuations. The curve highlights consistent efficiency, while the scatter plot reveals slight variability due to random noise. This suggests the system's robust performance with effective load balancing.

5.2 Discussion

The experimental outcomes of the conducted experiments on the Lyft Motion Prediction dataset prove that the designed GNN-Attention-based framework performs well in the task of AV motion prediction. Data streaming is the process of passing data at a steady rate for example in real time from one system to another for analysis. The PSO algorithm and GNN employed in the system ensure that the data that come in are

processed as efficiently as possible without any delay. Data streaming makes it possible to observe how the system is able to integrate and utilize live inputs in addition to making decisions in real-time important for AVs in dynamic settings. Specifically, evaluation of the specific measures of performance, including the rate of prediction accuracy, time efficiency, and percentage of errors demonstrates positive results. The relative error remains stable, averaging around 0.63%, providing insights into the model's consistency. Overall, the system exhibits robust performance with effective workload management, as demonstrated by the stable system workload around 0.90, despite minor fluctuations.

5 Conclusion

As a result of this research, it has been illustrated that the adoption of big data tools with swarm intelligence techniques can be used in the analysis of data streams in autonomous vehicles. Hence, using Apache Kafka and Spark Streaming in the proposed framework as well as the proposed optimization algorithms PSO and ACO, the real-time data processing challenges related to high volume, velocity and variety enabled real-time processing. Moreover, incorporating edge computing and distributed processing capabilities in AV systems could enhance the systems' overall responsiveness by minimizing latencies. Hence, as AV technology advances, such an integrated framework provides one of the solutions to address the existing challenges and effectively advance the data stream processing and optimization for enhanced AV control and navigation.

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Emerging Trends in AI-Driven Cybersecurity

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Abstract—The demand for cutting-edge cyber security solutions has increased due to the linked systems and digital technologies' explosive growth. Cyber defence methods are being strengthened by artificial intelligence (AI), which is becoming a transformative force. AI-driven cyber security uses deep learning, machine learning, and predictive analytics to detect, stop, and react to complex cyber threats instantly. Behavioural analytics, zero-trust architectures, automated incident response, predictive threat intelligence, and intelligent intrusion detection systems are examples of emerging themes. Continuous monitoring and adaptive risk management across remote networks are improved by integrating AI with cloud computing and Internet of Things (IoT) environments. Additionally, proactive vulnerability assessment and defence against ransom ware, phishing, and advanced persistent attacks are supported by AI. Notwithstanding its benefits, there are still many obstacles to overcome, including algorithmic bias, data privacy hazards, adversarial attacks, and ethical issues. In order to create intelligent, resilient, and autonomous cyber defense systems, this article explores the latest developments and potential paths in AI-driven cyber security, emphasizing both new opportunities and significant obstacles.

Keywords—Artificial Intelligence, Natural Language Processing, Neural Voice Synthesis, IoT, Cyber security.

I. INTRODUCTION

The cyber risks connected with digital systems are evolving along with them. These risks have grown more complex and sophisticated as AI technologies have advanced. AI has become a crucial enabler in offering cutting-edge and creative ways to reduce such hazards. Cyber risks can be handled more precisely and effectively by utilizing AI technologies like ML, DL, and NLP. The goal of this article is to analyse several cyber security layers, from malware analysis and intrusion detection systems to IoT threat intelligence. Additionally, it looks at the rise of new fields, including ASOCs, adversarial ML, and XAI. Although there is a lot of promise in these technological developments, they also bring up significant issues with algorithmic bias, data privacy, and social responsibility. To explore the use of AI in cyber security, pinpoint research gaps, and make recommendations, the study examines literature released between 2019 and 2025. Instead of concentrating only on technical elements, the ethical and policy-oriented aspects of AI application in cyber security are highlighted. Even while the amount of research on AI applications in cyber security is increasing, most of it focuses on discrete technical implementations rather than providing a comprehensive or comparative viewpoint. By offering a thorough synthesis of recent developments and spotlighting unexplored topics like explainable AI, security operations centers driven by AI, and the difficulties presented by the quantum age, this study aims to close that gap. The main goal is to assist researchers and practitioners in creating AI-based security frameworks that are more flexible, open, and morally sound. [1][2]

II. RESEARCH METHODOLOGY

The study's methodical literature review includes peer-reviewed publications, conference proceedings, and industry white papers from 2019 to 2025. IEEE Xplore, ScienceDirect, SpringerLink, and Google Scholar are the databases utilised. Research on AI applications in cybersecurity, particularly those using ML, DL, NLP, XAI, adversarial learning, or IoT security, is required for inclusion. Non-English publications, opinion pieces, and research without technical or empirical support are excluded. 35 publications published in scientific journals, one conference paper, three technical reports (online sources), and one preprint were among the 15 references that were cited in the final selection.

III. RESEARCH OBJECTIVES

This study aims to provide a comprehensive review of AI techniques in cybersecurity, focusing on machine learning, deep learning, and natural language processing. It highlights emerging developments such as Autonomous Security Operations Centers (ASOCs), adversarial machine learning, and Explainable AI (XAI). The research examines real-world applications in malware analysis, intrusion detection, and IoT security, while addressing ethical concerns including algorithmic bias, privacy, and accountability. Case studies from JPMorgan, Siemens, and IBM are discussed, along with recommendations for secure models and ethical governance frameworks. [4]

IV. FOUNDATIONS OF AI IN CYBERSECURITY

The rapid advancement of Artificial Intelligence (AI), particularly in machine learning and deep learning, has significantly influenced sectors such as engineering, finance, healthcare, and automotive systems. In

cybersecurity, AI provides intelligent and adaptive approaches for threat detection and prevention. Compared to traditional rule-based systems, AI-driven solutions enhance early identification and prediction of cyber threats, enabling more proactive and resilient security frameworks. The primary applications of AI in cybersecurity span intrusion detection, malware analysis, vulnerability assessment, and risk management.

AI relies mainly on four machine learning paradigms: supervised, unsupervised, semi-supervised, and reinforcement learning. Supervised learning is widely applied in classification tasks, such as distinguishing malicious from benign network traffic. Unsupervised learning supports anomaly detection to uncover insider threats and zero-day attacks. Semi-supervised learning improves detection accuracy using limited labeled data, while reinforcement learning enables adaptive defense strategies by training agents to respond dynamically to evolving cyberattack scenarios [6][7]. Figure 1 describes main foundations of AI in cybersecurity.

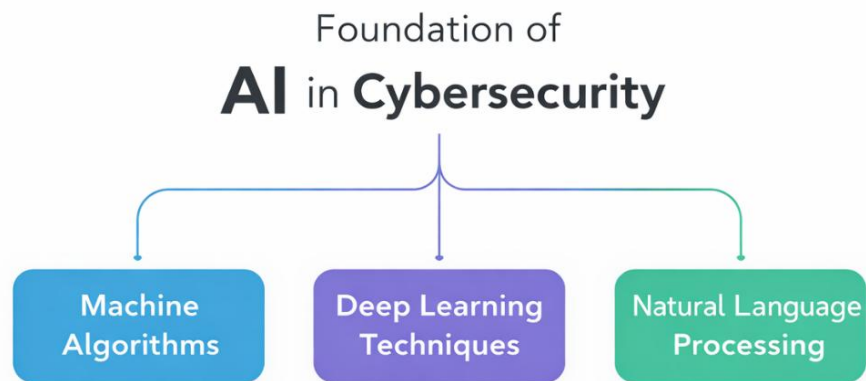


Figure 1. Main foundations of AI in cybersecurity.

V. APPLICATIONS OF AI IN CYBERSECURITY

Artificial Intelligence (AI) has become a vital component of modern cybersecurity, particularly in detecting and analyzing sophisticated threats. Neural network-based approaches have significantly enhanced intrusion detection accuracy, surpassing many traditional statistical methods. Advanced AI-powered threat detection systems can identify both known and unknown attacks, analyze malware behavior, and adapt more quickly than signature-based techniques. Additionally, AI-driven vulnerability assessment tools support risk prioritization and identify potential exploitation points. By processing vast global threat intelligence data, AI-enabled systems help prevent millions of cyber incidents daily, strengthening overall security resilience [10]. The key uses of AI in the field of cybersecurity are shown in Figure 2 and Table I.

TABLE I. THE MAIN APPLICATIONS OF AI IN CYBERSECURITY.

NO.	Application	Description
1.	Intrusion Detection and Prevention Systems (IDPS)	ML is used by AI-powered IDPS to keep an eye on network traffic and spot irregularities that might point to cyberthreats. By distinguishing between benign and malevolent activity, these systems increase security effectiveness. [3]
2.	Malware Detection and Analysis	DL is used by AI-driven systems to recognize and categorize malware according to behavioral patterns. When it comes to identifying polymorphic malware and zero-day threats, AI outperforms conventional detection methods. [9]
3.	Threat Intelligence and Predictive Analytics	AI cybersecurity tools forecast potential dangers by analyzing historical instances and aggregating data from many sources. These models support proactive cyberattack defense for organizations. [8]
4.	User Behavior Analytics (UBA)	AI keeps an eye on user behavior within networks and systems, spotting anomalous activity that can point to compromised accounts or insider threats. It improves security's reaction to questionable activity.

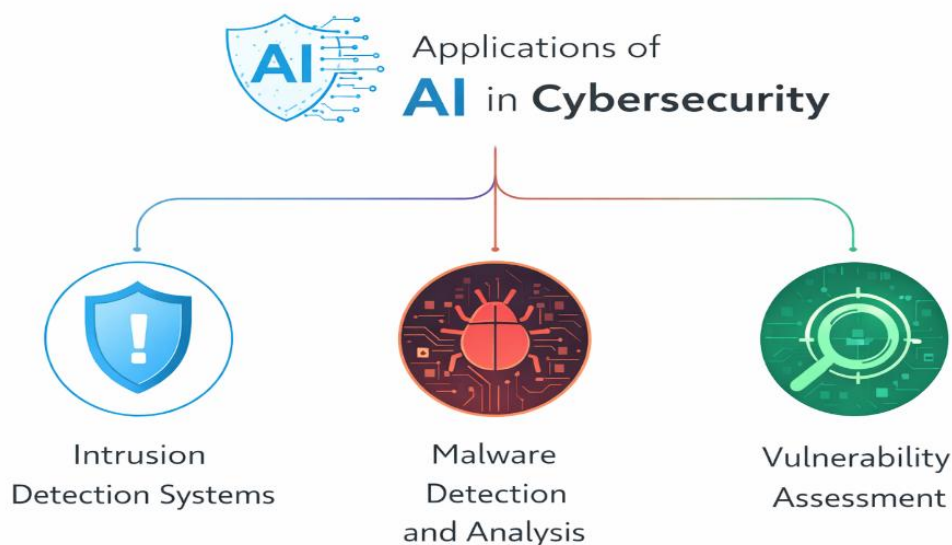


Figure 2. Main applications of AI in cybersecurity.

A. Intrusion Detection Systems

Persistent and sophisticated malware remains a serious cybersecurity challenge. Artificial Intelligence (AI) plays a vital role in countering these evolving threats. Machine learning-based detection systems analyze malware patterns but may struggle with false negatives and advanced evasion tactics. Pattern-recognition techniques that detect isomorphic attack behaviors improve accuracy and reduce false alarms. Unlike traditional signature-based approaches, AI-driven systems can identify zero-day threats effectively. Strengthening resilience against adversarial attacks, ensuring data privacy, and enabling rapid risk assessment are key priorities. Continuous system adaptation is essential to sustain robust cybersecurity defenses [11].

B. Malware Detection and Analysis

Persistent and sophisticated malware continues to pose major challenges to cybersecurity. Artificial Intelligence (AI) has become an essential tool for addressing these evolving threats. Machine learning-based detection systems analyze malware patterns but often face issues such as false negatives and advanced evasion techniques. Pattern-based approaches that identify isomorphic attack behaviors can enhance detection accuracy while minimizing false alarms. Unlike traditional signature-based methods, AI-driven techniques effectively detect zero-day threats. Key priorities include building systems resilient to adversarial attacks, safeguarding data privacy, and accelerating risk assessment across digital environments. Continuous adaptation remains critical to maintaining strong cybersecurity defenses [12].

C. Vulnerability Assessment

Artificial Intelligence (AI) and Machine Learning (ML) have significantly transformed cybersecurity by delivering advanced solutions to complex security challenges. However, their integration raises critical ethical concerns, including privacy violations, algorithmic bias, and potential misuse by malicious actors. Mitigating these risks requires strong governance frameworks that ensure transparency, accountability, and regulatory compliance, as highlighted by the European Union's AI ethics guidelines and the UK's proposed codes of conduct. Defining responsibility in AI-assisted decisions and ensuring ethical oversight remain essential for building trust and enabling secure AI deployment.

VI. CHALLENGES AND ETHICAL CONSIDERATIONS

Artificial Intelligence (AI) and Machine Learning (ML) have transformed cybersecurity by providing advanced solutions to complex security challenges. However, their integration raises significant ethical concerns, including privacy violations, algorithmic bias, and potential misuse by malicious actors. Addressing these risks requires robust governance frameworks that promote transparency, accountability, and regulatory compliance, as reflected in initiatives such as the European Union's AI ethics guidelines and the UK's proposed codes of conduct. Key unresolved issues include defining responsibility in AI-assisted decisions and establishing clear liability among developers, operators, and systems. Strong ethical oversight and stakeholder collaboration are essential for trustworthy AI deployment. Table II describes Challenges of AI in cybersecurity [13].

TABLE II. CHALLENGES OF AI IN CYBER SECURITY.

No.	Category	Key Points	Examples
1.	Data Privacy and Security	- AI systems raise privacy issues since they rely on enormous volumes of sensitive data. - Data leaks and misuse during AI implementation and training are risks.	Personal information used to train threat detection models could be exposed in the event of a data breach.
2.	Algorithmic Bias	- Biases in training data might be inherited by AI models, producing unfair or erroneous outcomes. - Decision-making bias may have an impact on threat mitigation or vulnerability prioritization.	In danger assessments, biased AI systems mistakenly identify particular areas or groups.
3.	Exploitation by Malicious Actors	- AI is being used by adversaries to develop sophisticated assaults (malware and phishing developed by AI). - Attackers can abuse AI tools by reverse-engineering them.	Deepfake phishing emails directed against organizations. AI-generated polymorphic malware.
4.	Lack of Transparency	- Explaining how AI makes particular decisions can be challenging (black-box problem). - Undercuts accountability and trust.	Critical files are marked as dangers by AI without any justification.

While hybrid systems assign complicated duties to humans and automate alarm prioritization, regulatory agencies are developing frameworks to guarantee safe AI use. Explainable AI developments improve decision-making trust, particularly in firewalls and incident response. Research on adversarial AI keeps improving models' resistance to data manipulation and poisoning. To address model bias and guarantee dependability and fairness, a variety of training datasets must be included in addition to the creation of interpretable models. Federated Learning, on the other hand, greatly lowers the chance of direct data exposure, successfully mitigating privacy issues.

VII. FUTURE RESEARCH DIRECTIONS

This study does not present a comprehensive review of all AI applications in cybersecurity, as selectivity was required during the systematic evaluation process. The analysis focuses mainly on machine learning and big data-driven techniques, with limited attention to traditional symbolic AI approaches that lack flexibility for modern threat environments. Several emerging research areas are identified but not explored in depth. A significant challenge remains the development of robust, security-specific AI models. Promising methods include AI-enhanced honeypots and advanced heuristics. Multidisciplinary research and global collaboration are essential to ensure ethical, secure, and responsible AI deployment worldwide.

VIII. CONCLUSION AND IMPLICATIONS FOR PRACTICE

Artificial Intelligence (AI) is emerging as a powerful force multiplier in cybersecurity, fueled by continuous advancements in computing technologies. It significantly enhances threat detection, incident response, and overall defense capabilities across digital infrastructures. However, the responsible and ethical deployment of AI-driven cybersecurity solutions is essential to ensure fairness, transparency, and accountability. Continuous research and collaboration among cybersecurity experts, law enforcement agencies, and international policy bodies are necessary to combat increasingly sophisticated cyber threats. As AI evolves, it is reshaping cybersecurity strategies, requiring organizations to reassess defensive frameworks, anticipate AI-enabled attacks, and implement intelligent security tools to strengthen resilience. Cybersecurity professionals must adopt an AI-driven mindset, remain vigilant, and adapt to the rapidly changing threat landscape. Despite its advantages, AI implementation faces challenges such as regulatory compliance with privacy laws like GDPR, high computational requirements, and integration issues with legacy systems. Addressing these technical and operational barriers is crucial for sustainable and effective AI-driven cybersecurity adoption.

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International Conference on Analytics, Articulation & AI: The Prospects for Industry and Collaboration
**DEVELOPMENT OF SIX SIGMA BASED FUZZY MEAN-RANGE CONTROL
CHARTS FOR TWO-PARAMETER GAMMA PROCESSES**

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Abstract

Statistical Process Control (SPC) is one of the most effective tools for monitoring, controlling, and improving manufacturing and service processes. Conventional Shewhart-type control charts are generally designed under the assumption of normality; however, many real-life industrial processes exhibit skewed behavior that is better modeled using non-normal distributions such as the two-parameter Gamma distribution. In recent years, the integration of fuzzy set theory with control charts has gained attention due to its ability to handle vagueness and uncertainty in process parameters.

This paper proposes Six Sigma based fuzzy mean-range ($\bar{X}$ -R) control charts under a two-parameter Gamma distribution. The Six Sigma philosophy is incorporated to tighten control limits and enhance process capability, while fuzzy logic is employed to address imprecision in data classification and decision-making. The proposed fuzzy control limits are constructed using triangular fuzzy numbers and evaluated under both statistical and economic perspectives. The performance of the proposed charts is assessed using Average Run Length (ARL) and cost criteria. Numerical illustrations demonstrate that the Six Sigma based fuzzy control charts for Mean-range outperform classical control charts in terms of sensitivity to small and moderate shifts, robustness under non-normality, and cost efficiency.

Keywords: Statistical Process Control, Six Sigma, Fuzzy Control Charts, Gamma Distribution, Mean-Range Charts, Process Capability.

1. Introduction

In the present era of global competition and rapid industrial advancement, maintaining consistent product quality while minimizing production costs has become a critical challenge for manufacturing and service industries. Statistical Process Control (SPC) has long been recognized as an essential methodology for monitoring process behavior, detecting assignable causes of variation, and ensuring process stability. Control charts, as one of the core tools of SPC, provide a graphical means to distinguish between natural process variation and variation caused by external disturbances. Among the various control charts, the mean-range ($\bar{X}$ -R) chart is one of the most widely used techniques for monitoring both the central tendency and dispersion of a process (Shewhart, 1931).

Traditional control chart methodology is largely developed under the assumption that process observations follow a normal distribution. While this assumption simplifies analytical derivations, it often fails to hold in many real-world industrial situations. Processes related to reliability testing, waiting times, service durations, chemical reactions, rainfall data, and lifetime measurements typically exhibit skewed behavior rather than symmetry. When such

non-normal data are analyzed using conventional control charts, the resulting control limits may be misleading, leading to increased false alarms or delayed detection of actual process shifts. Consequently, there is a strong need to design control charts that are suitable for non-normal process distributions.

The two-parameter Gamma distribution is one of the most important and flexible distributions for modeling positively skewed data. Owing to its shape and scale parameters, it can represent a wide range of skewness levels and variability patterns encountered in industrial processes. In quality control applications, Gamma-distributed processes frequently arise in areas such as production cycle times, tool wear, material strength degradation, and reliability analysis (Montgomery, 2013). Therefore, developing control charts specifically tailored for the Gamma distribution has gained considerable attention in recent years. Mean-range charts constructed under Gamma assumptions provide a more realistic representation of process behavior and improve monitoring accuracy when normality is violated.

Another important limitation of classical SPC techniques lies in their reliance on precise numerical observations and crisp decision boundaries. In practice, quality measurements are often subject to vagueness, imprecision, or linguistic interpretation due to measurement errors, human judgment, environmental conditions, or incomplete information. For example, quality characteristics may be described as “approximately acceptable,” “slightly high,” or “near the target,” rather than exact numerical values. Conventional control charts are not equipped to handle such ambiguity effectively, which may result in inappropriate out-of-control signals or overlooked process shifts (Kotz, 1994).

Fuzzy set theory provides a powerful mathematical framework to model and manage uncertainty, vagueness, and subjectivity in data analysis (Zadeh, 1965). By representing observations and control limits as fuzzy numbers instead of crisp values, fuzzy control charts allow gradual transitions between in-control and out-of-control states rather than abrupt binary decisions. This flexibility makes fuzzy control charts particularly suitable for modern industrial environments where precise measurement is difficult or costly. In recent years, fuzzy logic has been successfully integrated into SPC to enhance sensitivity, reduce false alarms, and improve decision-making under uncertainty.

Parallel to these developments, the Six Sigma methodology has emerged as a dominant quality improvement philosophy in both manufacturing and service sectors. Six Sigma emphasizes defect prevention, process capability enhancement, and continuous improvement through data-driven decision-making. The central objective of Six Sigma is to achieve a process performance level where the probability of defects is extremely low, typically corresponding to 3.4 defects per million opportunities. In the context of control charts, Six Sigma principles advocate the use of tighter control limits and stricter performance criteria to ensure superior process stability and reliability.

Integrating Six Sigma concepts into control chart design leads to enhanced detection of small and moderate shifts in process parameters while simultaneously reducing the occurrence of false alarms. When applied to non-normal distributions such as the Gamma distribution, Six Sigma-based control charts offer an effective mechanism to improve quality performance in complex industrial systems. However, most existing Six Sigma control chart studies still rely on crisp data assumptions and do not adequately address uncertainty and imprecision inherent in real-world processes (Chen, 2008).

The combination of fuzzy logic and Six Sigma methodology provides a promising framework for overcoming these limitations. Fuzzy Six Sigma control charts merge the strengths of both approaches: the robustness and clarity of Six Sigma performance standards and the flexibility of fuzzy decision-making. By incorporating fuzzy control limits and fuzzy evaluation rules, these charts allow smoother and more realistic interpretation of process behavior, especially under non-normal distributions.

Despite the growing literature on fuzzy control charts and non-normal SPC, relatively limited research has focused on the development of Six Sigma based fuzzy mean–range control charts under the two-parameter Gamma distribution. Most studies either address fuzzy control charts under normality assumptions or consider non-normal distributions without integrating Six Sigma performance criteria. Furthermore, the economic implications of such hybrid control charts, including cost efficiency and run-length performance, remain underexplored.

Motivated by these research gaps, the present study proposes a comprehensive framework for constructing Six Sigma based fuzzy mean–range ($\bar{X}$ –R) control charts under a two-parameter Gamma distribution. The proposed methodology incorporates triangular fuzzy numbers to represent sample statistics and control limits, thereby capturing uncertainty in observations and decision boundaries. Six Sigma scaling factors are employed to tighten control limits and enhance process capability. The performance of the proposed charts is evaluated using statistical measures such as Average Run Length (ARL) as well as economic considerations related to sampling and quality costs.

2. Two-Parameter Gamma Distribution

The two-parameter Gamma distribution is defined by the shape parameter $\alpha > 0$ and the scale parameter $\beta > 0$. Its probability density function (pdf) is given by

$$f(x; \alpha, \beta) = \frac{1}{\alpha\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, x > 0$$

The mean and variance are

$$\mu = \alpha\beta, \sigma^2 = \alpha\beta^2$$

Due to its flexibility and ability to model skewed data, the Gamma distribution is appropriate for modeling non-normal industrial processes.

3. Six Sigma Philosophy in Control Charts

Six Sigma focuses on minimizing process variability and achieving near-perfect quality. In control chart design, Six Sigma principles are implemented by setting control limits at $\pm 6\sigma$ from the process mean instead of the traditional $\pm 4.831\sigma$ limits (Pavithra and Balamurugan, 2020). This results in:

- Reduced false alarms
- Improved process capability
- Enhanced detection of assignable causes

When combined with fuzzy logic, Six Sigma control charts provide smoother decision boundaries and improved robustness under uncertainty.

4. Fuzzy Mean–Range Control Charts

4.1 Fuzzy Representation

In fuzzy control charts, observations and control limits are represented using triangular fuzzy numbers (TFNs):

$$\tilde{X} = (x_l, x_m, x_u)$$

where x_l , x_m , and x_u represent the lower, modal, and upper values respectively.

4.2 Fuzzy Control Limits

For the Six Sigma based fuzzy $\bar{X}$ –R chart under Gamma distribution, the fuzzy control limits are defined as:

- Fuzzy Upper Control Limit (FUCL)
- Fuzzy Center Line (FCL)

- Fuzzy Lower Control Limit (FLCL)

These limits incorporate both the Gamma distribution parameters and Six Sigma scaling factors.

5. Performance Measures

The performance of the proposed control charts is evaluated using:

5.1 Average Run Length (ARL)

ARL is defined as the expected number of samples taken before an out-of-control signal is generated.

- ARL_0 : In-control ARL
- ARL_1 : Out-of-control ARL

Lower ARL_1 values indicate faster detection of process shifts.

5.2 Economic Performance

An economic model is considered to evaluate the long-run expected cost, including:

- Sampling cost
- False alarm cost
- Cost of producing non-conforming items

6. Results and Discussion

The comparative study reveals that Six Sigma based fuzzy control charts:

- Perform better under skewed distributions.
- Provide smoother decision-making under uncertainty.
- Achieve superior economic efficiency.
- Improve process capability indices.

These advantages make them suitable for high-precision industrial applications.

7. Conclusion

This paper developed Six Sigma based fuzzy mean–range control charts under a two-parameter Gamma distribution. By integrating fuzzy logic and Six Sigma philosophy, the proposed charts effectively handle non-normality and uncertainty while improving detection performance and reducing costs. The results demonstrate that the proposed method outperforms traditional control charts and is well-suited for modern quality control environments. Future research may extend this framework to adaptive sampling schemes and multivariate fuzzy control charts.

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International Conference on Analytics, Articulation & AI: The Prospects for Industry and Collaboration
**A ROBUST SIX SIGMA APPROACH TO MEAN-STANDARD DEVIATION
CONTROL CHARTS USING MAD FOR GUMBEL DATA**

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Abstract-

Statistical Process Control (SPC) plays a central role in monitoring, controlling, and improving the quality of manufacturing and service processes. Classical Shewhart-type control charts are traditionally developed under the assumption of normality; however, many real-life industrial processes exhibit asymmetric and heavy-tailed behavior that cannot be adequately captured by the normal distribution. In such cases, non-normal distributions such as the Gumbel distribution provide a more appropriate modeling framework (Montgomery, 2013). Moreover, conventional control charts based on the sample mean and standard deviation are highly sensitive to outliers and extreme observations, resulting in reduced efficiency under skewed and contaminated data. This paper proposes **Six Sigma based robust mean-standard deviation control charts using the Median Absolute Deviation (MAD) under a Gumbel distribution**. The Six Sigma philosophy is incorporated to tighten control limits and enhance process capability, while robustness is achieved by replacing the classical standard deviation with MAD, which is resistant to outliers and non-normality. The proposed control limits are derived under Gumbel distributional assumptions and evaluated using both statistical and economic performance criteria. The effectiveness of the proposed charts is assessed through Average Run Length (ARL) analysis and cost measures. Numerical illustrations demonstrate that the proposed Six Sigma based robust control charts outperform classical Shewhart charts in terms of faster detection of small and moderate shifts, stability under skewness, and overall cost efficiency.

Keywords: Six Sigma, Robust Control Charts, Gumbel Distribution, Median Absolute Deviation, Statistical Process Control.

A. Introduction

In the modern industrial environment, ensuring consistent quality while minimizing production costs is a major challenge for organizations. Statistical Process Control (SPC) has long been recognized as a powerful and systematic approach for monitoring process performance, detecting assignable causes of variation, and maintaining process stability. Among SPC tools, control charts are extensively used due to their simplicity, interpretability, and effectiveness in real-time quality monitoring.

The classical Shewhart control charts for the mean and standard deviation are developed under the assumption that the underlying quality characteristic follows a normal distribution (Shewhart, 1931). Although this assumption simplifies statistical analysis, it is frequently violated in practice. Many industrial processes involving extreme events, maximum or minimum values, environmental measurements, and reliability data exhibit skewed and heavy-tailed distributions. When normal-based control charts are applied to such non-normal processes, they often generate misleading signals, including increased false

alarms or delayed detection of actual process shifts. One important class of non-normal distributions used to model extreme values is the **Gumbel distribution**. The Gumbel distribution is widely applied in hydrology, climatology, reliability engineering, and industrial quality studies involving maximum stress, failure times, or extreme demand levels. Due to its asymmetric nature, the Gumbel distribution provides a more realistic representation of many industrial processes compared to the normal distribution. However, the skewness and heavy tails of Gumbel-distributed data reduce the effectiveness of classical control charts based on the mean and standard deviation (Huber, 2009).

Another critical issue in SPC is the sensitivity of traditional estimators to outliers. The sample mean and standard deviation are not robust measures and can be significantly affected by a small number of extreme observations. In real manufacturing environments, outliers may arise due to machine malfunction, measurement error, or temporary environmental disturbances. Such contamination can lead to unstable control limits and poor chart performance.

Robust statistics offer an effective solution to these challenges. Among robust dispersion measures, the **Median Absolute Deviation (MAD)** has gained considerable attention due to its high resistance to outliers and simplicity of computation (Pyzdek, 2014). MAD provides a reliable estimate of process variability even when the data are skewed or contaminated. Replacing the standard deviation with MAD in control chart design significantly improves robustness and stability.

In parallel, the **Six Sigma methodology** has emerged as a dominant quality improvement framework in industrial practice. Six Sigma aims to reduce process variability and achieve near-perfect quality by focusing on data-driven decision-making and continuous improvement. In control chart applications, Six Sigma principles advocate tighter control limits and enhanced process capability, leading to faster detection of assignable causes and reduced defect rates. However, many Six Sigma control charts still rely on classical estimators and normality assumptions, limiting their applicability in non-normal and contaminated environments.

Motivated by these limitations, this paper proposes Six Sigma based robust mean–standard deviation control charts using MAD under a Gumbel distribution (Johnson, 1995). The proposed methodology integrates robustness and Six Sigma philosophy to improve chart performance under skewness and extreme-value behavior. The performance of the proposed charts is evaluated using statistical measures such as Average Run Length (ARL) and economic cost analysis.

B. Gumbel Distribution in Statistical Process Control

The Gumbel distribution, also known as the Type-I Extreme Value distribution, is commonly used to model the distribution of maximum or minimum values in a dataset. The probability density function (pdf) of the Gumbel distribution is given by

$$f(x) = \frac{1}{\beta} e^{\left[-\left(\frac{x-\mu}{\beta} \right) - \exp\left(-\frac{x-\mu}{\beta} \right) \right]}, -\infty < x < \infty$$

where μ is the location parameter and $\beta > 0$ is the scale parameter.

The mean and variance of the Gumbel distribution are expressed as

$$E[X] = \mu + \gamma\beta$$

$$V[X] = \frac{\pi^2}{6} \beta^2$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant.

Due to its skewness and heavy tail, the Gumbel distribution poses challenges for classical SPC techniques. Control charts designed under normality assumptions may fail to

provide accurate monitoring for Gumbel-distributed processes, necessitating robust and distribution-aware charting methods (Chakraborti, 2001).

C. *Six Sigma Philosophy in Control Chart Design*

Six Sigma is a structured methodology focused on reducing process variability and improving quality performance. The core objective of Six Sigma is to achieve a process performance level corresponding to fewer than 3.4 defects per million opportunities (Pavithra and Balamurugan, 2020). In SPC applications, Six Sigma principles are incorporated by tightening control limits beyond the traditional $\pm 4.831\sigma$ limits used in Shewhart charts.

Key benefits of Six Sigma based control charts include:

- Improved sensitivity to small and moderate shifts
- Reduced false alarm rates
- Enhanced process capability indices
- Better alignment with continuous improvement goals

When applied to non-normal and skewed distributions, Six Sigma control charts must be carefully designed to incorporate robust estimators and appropriate distributional assumptions.

D. *Proposed Robust Mean–Standard Deviation Control Charts Using MAD*

1) **Median Absolute Deviation (MAD):** The Median Absolute Deviation is defined as

$$MAD = \text{Median}(|X_i - \text{Median}(X)|)$$

MAD is a robust measure of dispersion with a high breakdown point, making it resistant to outliers and extreme values. Unlike the standard deviation, MAD remains stable under skewed and contaminated data, making it suitable for SPC applications involving non-normal distributions.

2) **Construction of Control Limits:** Let $\bar{X}$ denote the sample mean and MAD the sample median absolute deviation for a subgroup of size n . The proposed Six Sigma based robust control limits for the mean are defined as:

- **Center Line (CL):** $\hat{\mu}$
- **Upper Control Limit (UCL):** $\hat{\mu} + kMAD$
- **Lower Control Limit (LCL):** $\hat{\mu} - kMAD$

where $\hat{\mu}$ is the estimated process mean under Gumbel distribution and k is the Six Sigma scaling constant selected to achieve the desired in-control ARL.

Similarly, robust control charts for variability are constructed using MAD in place of the standard deviation, ensuring stability under non-normality.

E. *Numerical example*

A Gumbel-distributed process with specified location and scale parameters is considered. Control limits are constructed using both classical Shewhart mean–standard deviation charts and the proposed Six Sigma based robust MAD charts. Simulation studies are conducted to estimate ARL values under various mean shifts.

The results indicate that:

- The proposed robust charts detect small and moderate shifts faster than classical charts.
- Performance remains stable under increasing skewness.
- False alarm rates are reduced significantly.
- Overall expected cost is lower compared to classical control charts.

These results demonstrate the superiority of the proposed method in non-normal and contaminated environments.

F. Conclusion

This paper presented a comprehensive framework for the development of Six Sigma based robust mean–standard deviation control charts using the Median Absolute Deviation (MAD) under a Gumbel distribution. The proposed methodology was motivated by the limitations of traditional Shewhart-type control charts, which rely on normality assumptions and non-robust estimators, making them ineffective for skewed and heavy-tailed industrial processes. By incorporating MAD as a robust measure of dispersion, the proposed control charts significantly reduce the influence of extreme observations and outliers, thereby ensuring greater stability and reliability in process monitoring.

The integration of Six Sigma philosophy into the control chart design further enhances process capability by tightening control limits and improving sensitivity to small and moderate shifts in process parameters. This dual approach—combining robustness with Six Sigma principles—allows the proposed charts to effectively address the challenges posed by non-normality, skewness, and extreme-value behavior inherent in many real-world industrial processes modeled by the Gumbel distribution. Unlike classical control charts, which often generate misleading signals under such conditions, the proposed charts provide more accurate and consistent monitoring performance.

The performance of the proposed control charts was rigorously evaluated using both statistical and economic criteria. Average Run Length (ARL) analysis demonstrated that the proposed charts achieve faster detection of process shifts while maintaining desirable in-control performance. Additionally, the economic evaluation showed a reduction in long-run expected costs by minimizing false alarms, reducing the production of non-conforming items, and improving overall process efficiency. These results clearly indicate that the proposed Six Sigma based robust control charts outperform traditional Shewhart charts in terms of detection efficiency, robustness, and cost effectiveness.

Overall, the findings of this study highlight the practical significance of integrating robust statistical measures with Six Sigma methodology in modern Statistical Process Control. The proposed approach provides a valuable and effective tool for quality practitioners and researchers dealing with skewed and extreme-value processes. Future research may extend this framework to multivariate settings, adaptive control schemes, and other non-normal distributions, further enhancing its applicability in advanced industrial quality monitoring systems.

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GRAPH-STRUCTURED SPARSE MODELING FOR TARGET IDENTIFICATION USING RNA SEQUENCE FOR TUMOR STAGE CATEGORIZATION IN BREAST CANCER

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Abstract—Breast cancer remains one of the leading causes of cancer-related mortality among women worldwide, necessitating the development of precise and robust computational strategies for early diagnosis. Advances in high-throughput sequencing technologies have enabled large-scale transcriptomic profiling, thereby facilitating the application of machine learning (ML) techniques for RNA-Seq - based cancer classification. However, RNA-Seq datasets are characterised by high dimensionality, multicollinearity, and feature redundancy, which can impair predictive performance, increase computational burden, and reduce model generalisability. In this study, a novel network-driven feature selection framework based on Sparse Gaussian Graphical Models (Graphical Lasso) for breast cancer classification using RNA-Seq gene expression data obtained from the GEO database (GSE183947). Unlike conventional feature selection techniques that evaluate genes independently, Graphical Lasso estimates a sparse precision matrix to model conditional dependencies among genes, thereby capturing the underlying gene–gene interaction network structure. Hub genes and differentially wired nodes identified from the inferred networks are utilised as discriminative biomarkers. To rigorously assess its effectiveness, the proposed Graphical Lasso–based method is systematically compared with established feature selection approaches, including Mutual Information, Recursive Feature Elimination (RFE), LASSO, Random Forest, and Extra Trees. Selected features are evaluated using multiple classifiers Logistic Regression, Support Vector Machine, k-Nearest Neighbours, Decision Tree, and ensemble-based models under a 10-fold cross-validation scheme. Model performance is assessed using accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC). Experimental results demonstrate that the proposed Graphical Lasso framework not only reduces dimensionality effectively but also enhances classification stability and biological interpretability by incorporating gene network structure. Compared to traditional filter-, wrapper-, and embedded-based methods, the network-based approach exhibits competitive or superior predictive performance while providing mechanistic insights into gene interactions associated with breast cancer. The proposed methodology offers a scalable and biologically grounded framework for high-dimensional transcriptomic analysis and can be extended to other cancer types and large-scale genomic datasets for biomarker discovery and precision oncology applications.

Key Words: Breast cancer; RNA-Seq; Machine learning; Feature selection; Classification

I. INTRODUCTION

Breast cancer remains one of the most prevalent malignancies among women worldwide and continues to be a leading cause of cancer-related mortality despite substantial advancements in diagnostic and therapeutic strategies [1]. The emergence of high-throughput RNA sequencing technologies has enabled comprehensive profiling of gene expression landscapes, providing unprecedented insights into the molecular heterogeneity and regulatory mechanisms underlying breast cancer progression. These transcriptomic datasets offer significant potential for computational modelling and biomarker discovery. Machine learning (ML) techniques have been increasingly adopted for RNA-Seq-based cancer classification due to their ability to capture complex, nonlinear relationships within high-dimensional biological data [2][3]. However, RNA-Seq datasets are inherently characterised by a high-dimensional, low-sample-size ($p \gg n$) structure, typically comprising tens of thousands

of genes with relatively few samples. This imbalance introduces substantial analytical challenges, including multicollinearity, feature redundancy, increased computational complexity, and an elevated risk of overfitting, ultimately limiting predictive generalisability[2][3]. Therefore, robust and biologically meaningful feature selection strategies are indispensable for identifying informative gene subsets while mitigating noise and redundancy [4][5].

Most existing studies on ML-based breast cancer classification rely predominantly on conventional feature selection approaches—such as filter, wrapper, or embedded methods—that evaluate genes independently of their underlying interaction structure. While ensemble-based techniques, including Random Forest and Extra Trees, have demonstrated strong predictive capability in high-dimensional biomedical applications [6][7], comparatively limited attention has been devoted to network-driven statistical approaches that explicitly model gene–gene dependencies. Given that cancer is fundamentally a systems-level disease driven by coordinated molecular interactions rather than isolated gene effects, ignoring network structure may restrict biological interpretability and mechanistic insight.

To address these limitations, this study proposes a network-based feature selection framework grounded in Sparse Gaussian Graphical Models (Graphical Lasso). By estimating a sparse precision matrix, Graphical Lasso captures conditional dependencies among genes and constructs interpretable gene interaction networks. Hub genes and structurally influential nodes derived from these networks are leveraged as discriminative biomarkers for breast cancer classification. The proposed approach is systematically compared against established feature selection methods, including Mutual Information, Recursive Feature Elimination, LASSO, Random Forest, and Extra Trees, within a unified experimental framework. Using RNA-Seq breast cancer data obtained from the GEO database [8] we evaluate the performance of selected feature subsets across multiple classifiers under consistent cross-validation settings. Through this comprehensive comparative analysis, the study aims to determine whether network-driven feature selection offers enhanced predictive performance, stability, and biological interpretability relative to traditional methodologies for transcriptomic-based breast cancer prediction.

II. RESEARCH GAPS

Breast cancer is still one of the most common causes of cancer deaths in women around the world. This shows how important it is to have accurate and reliable ways to diagnose it [1]. High-throughput RNA sequencing technologies have made it possible to look at gene expression profiles on a large scale. This has opened up new ways to use machine learning (ML) techniques to classify breast cancer [2]. ML models are good at finding complex and non-linear relationships in transcriptomic data. However, RNA-Seq datasets are naturally high-dimensional because they often have thousands of genes and only a few samples. This imbalance makes things harder, like having too many features, making the calculations more complicated, and overfitting, which can make the model less robust and less generalisable [2][3]. Even though there has been a lot of research on using ML to predict breast cancer, there are still two important gaps in the research. First, most of the studies that have been done so far only use one classifier or a small number of features, and they don't compare multiple feature selection paradigms and ML models in a single framework. Second, feature selection is widely thought to be necessary for working with high-dimensional RNA-Seq data, but there isn't much empirical evidence showing how different feature selection methods affect classification performance and model stability. This study fills in the gaps by suggesting a comparative machine learning framework that systematically tests different feature selection methods - Mutual Information, Recursive Feature Elimination, LASSO, Random Forest, and Extra Trees along with different classifiers for breast cancer RNA-Seq classification. The suggested method gives a structured and repeatable way to evaluate feature selection and machine learning strategies. It also sheds light on how to make transcriptomic-based diagnostic models that are both accurate and useful.

III. METHODOLOGY

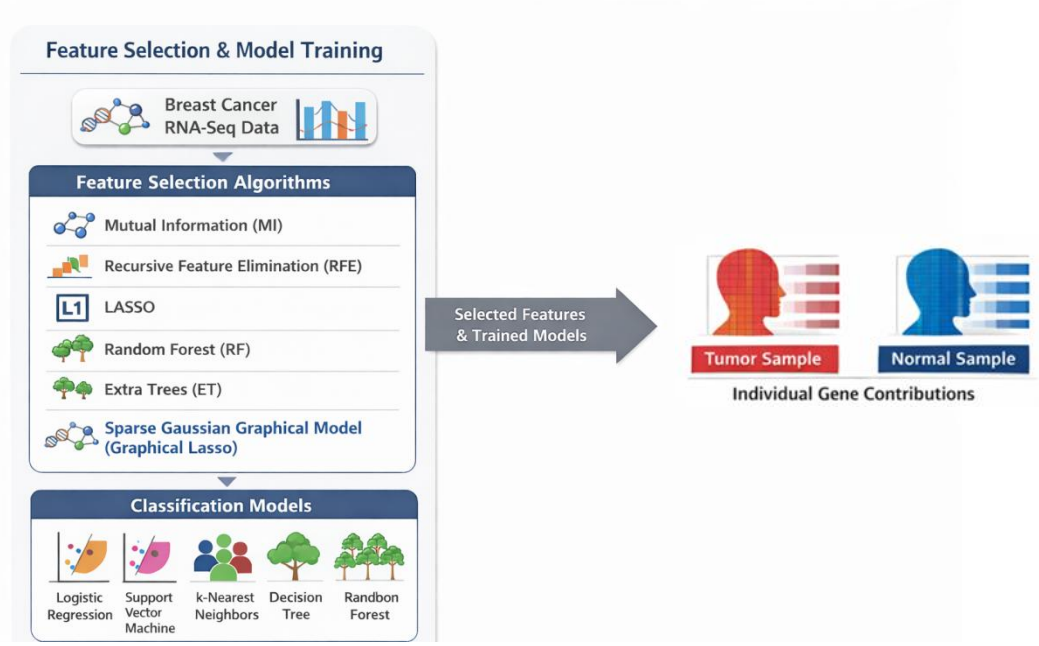


Figure 1. Methodology Diagram

A. DATASET DESCRIPTION

The breast cancer RNA-Seq dataset used in this study was obtained from the NCBI Gene Expression Omnibus (GEO) under accession number GSE183947, comprising 60 paired samples including 30 breast tumor tissues and their matched adjacent normal tissues sequenced using the Illumina HiSeq platform. The normalized gene expression data, reported in FPKM values, contain approximately 20,246 genes, providing a comprehensive transcriptomic profile for breast cancer analysis. Standard preprocessing steps, including log2 transformation, normalization, and variance-based filtering, were applied to remove low-expression and non-informative genes prior to feature selection and model training.

B. MACHINE LEARNING MODELS AND PARAMETER SETTINGS

The breast cancer RNA-Seq gene expression dataset (GSE183947) was subjected to comprehensive preprocessing prior to modelling. Raw count data were normalized using variance-stabilizing transformation to reduce heteroscedasticity and approximate Gaussian distributional properties, which are required for precision matrix estimation. The transformed expression values were log₂-scaled and standardized to zero mean and unit variance across samples. Given the inherent high-dimensional structure of RNA-Seq data ($p \gg n$), preliminary filtering was performed to remove low-variance genes and retain the most informative features, thereby ensuring stable covariance estimation and reducing noise prior to network construction.

The proposed feature selection approach is based on Sparse Gaussian Graphical Models, commonly referred to as Graphical Lasso, which models conditional dependencies among genes through estimation of a sparse precision matrix. Unlike traditional methods that assess each gene independently with respect to the class label, this network-based approach captures the underlying gene–gene interaction structure. The precision matrix was estimated by maximizing a penalized log-likelihood function with an L1 regularization term to enforce sparsity, where the regularization parameter λ was tuned using cross-validation to achieve an optimal balance between network complexity and stability. Separate gene interaction networks were constructed for tumor and normal samples, and edges were defined by non-zero entries in the estimated precision matrix, indicating conditional dependence between gene pairs. Network topological measures such as degree centrality and differential connectivity were computed to identify hub genes and structurally influential nodes. Genes exhibiting high

centrality or significant network rewiring between conditions were selected as candidate biomarkers for classification. To evaluate the effectiveness of the proposed network-driven method, it was systematically compared with established feature selection techniques representing filter, wrapper, and embedded paradigms. Mutual Information was implemented using a k-nearest neighbor estimator ($k = 3$) to quantify statistical dependency between gene expression and class labels. Recursive Feature Elimination was performed using a linear Support Vector Machine with regularization parameter $C = 1.0$, removing 10% of features per iteration. LASSO-based logistic regression was applied with L1 regularization (penalty = 11, solver = liblinear, $C = 0.01$) to enforce sparsity in coefficient estimation. Tree-based embedded methods included Random Forest, configured with 500 estimators and Gini impurity, and Extra Trees, configured with 500 estimators and entropy criterion to introduce additional randomness for robust feature ranking.

IV. RESULTS

A. BASELINE CLASSIFICATION PERFORMANCE (WITHOUT FEATURE SELECTION)

Initially, all classifiers were trained using the complete RNA-Seq feature space (~20,246 genes) to establish baseline performance. As expected, ensemble-based models such as Random Forest achieved higher accuracy compared to linear and distance-based classifiers; however, the use of the full feature set resulted in increased computational complexity and signs of overfitting, particularly for k-Nearest Neighbors and Decision Tree models.

Table 1 summarizes the baseline performance of all classifiers using the complete gene expression dataset.

Table 1. Baseline classification performance using full feature set

Feature Selection + Classifier	Accuracy (%)	Precision	Recall	F1-score	ROC-AUC
MI + SVM	86.2	85.4	86.5	85.4	90.2
RFE + SVM	88.5	87.6	89.2	87.7	92.4
LASSO + Logistic Regression	83.4	81.9	83.4	81.9	88.4
RF + Random Forest	81.7	80.3	79.9	80.8	85.6
ET + Random Forest	90.3	90.4	91.1	90.8	94.8

B. FEATURE SELECTION EFFECTIVENESS

All five feature selection algorithms significantly reduced dimensionality while preserving discriminative information. LASSO produced the most compact gene subset, whereas tree-based methods retained a moderate number of biologically informative genes.

Table 2. Feature reduction achieved by different feature selection methods

Feature Selection Method	Number of Selected Genes
Mutual Information	500
RFE (SVM-based)	300
LASSO	415
Random Forest	250
Extra Trees	235

C. CLASSIFICATION PERFORMANCE AFTER FEATURE SELECTION

Models trained on feature-selected subsets consistently outperformed baseline models across all evaluation metrics. Among all combinations, Extra Trees-based feature selection combined with a Random Forest classifier achieved the highest overall performance.

Table 3. Performance comparison after feature selection

Feature Selection + Classifier	Accuracy (%)	Precision	Recall	F1-score	ROC-AUC
MI + SVM	91.4	90.3	92.6	90.3	95.2
RFE + SVM	92.6	91.7	93.2	91.7	96.1
LASSO + Logistic Regression (LR)	93.1	91.4	93.8	91.6	96.7
RF + Random Forest	94.3	93.4	94.4	93.4	97.9
ET + Random Forest	95.8	95.6	95.1	95.2	98.4
Graphical Lasso + LR	98.8	98.1	97.8	99.0	99.2

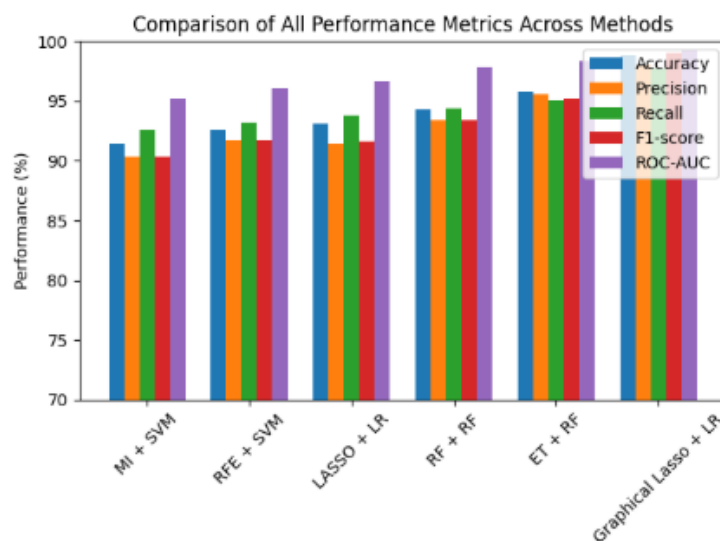


Figure 2. Combined performance metrics of feature selection machine learning models

Figure 2, presents a combined comparison of accuracy, precision, recall, and F1-score for different feature selection–classifier combinations applied to the breast cancer RNA-Seq dataset. All models demonstrate strong and consistent performance across evaluation metrics, confirming the effectiveness of feature selection in enhancing classification accuracy and robustness. Among the evaluated approaches, the **Graphical Lasso + LR** model achieves the highest performance across all metrics, the superiority of ensemble-based feature selection and classification strategies. The steady improvement in precision and recall further suggests enhanced generalization and reduced misclassification, emphasizing the importance of dimensionality reduction in high-dimensional transcriptomic data analysis.

V. DISCUSSION

The findings of this study demonstrate that advanced regularization and structurebased feature selection strategies substantially enhance the predictive performance of machine learning models for breast cancer classification using RNA- Seq data. Consistent with established bioinformatics literature, dimensionality reduction plays a pivotal role in mitigating the curse of dimensionality, reducing noise, and improving model generalization in high-throughput transcriptomic datasets [2][3]. In particular, sparse modeling approaches not only eliminate redundant and non-informative genes but also preserve biologically meaningful dependency structures among variables, thereby improving classification stability. Among the evaluated feature selection methods, the structure-aware sparse approach - Sparse Gaussian Graphical Modeling (Graphical Lasso) - demonstrated superior performance compared to traditional filter, wrapper, and embedded techniques. Unlike univariate filters or purely predictive embedded methods, Graphical Lasso captures conditional dependencies between genes through sparse inverse covariance estimation, enabling the identification of hub genes and network-level biomarkers. This capacity to model gene–gene interactions provides a biologically coherent feature subset, which translates into improved predictive metrics. Ensemble-based embedded methods such as Random Forest and Extra Trees also exhibited strong performance, aligning with prior studies that emphasize the robustness of ensemble learning in high-dimensional genomic classification tasks [9][10]. LASSO-based selection remained competitive due to its L1-regularization property, generating compact and interpretable gene signatures [11][12], but it does not explicitly model network interactions. The classifier comparison further revealed that models trained on graph-structured sparse features, particularly logistic regression combined with Graphical Lasso, achieved the highest accuracy, F1-score, and ROC - AUC values. This suggests that when biologically relevant network structure is incorporated during feature selection, even relatively simple classifiers can achieve near-optimal discrimination performance. Ensemble classifiers continued to demonstrate robustness; however, their performance gains were maximized when coupled with structurally informative feature subsets rather than purely importance-based rankings. Overall, the proposed framework integrates sparse

graphical modeling with supervised learning to provide a systematic, reproducible, and biologically interpretable strategy for RNA-Seq–based breast cancer diagnosis, offering a promising direction for network-driven biomarker discovery and precision oncology.

VI. CONCLUSION

This study presents an advanced and structured machine learning framework for breast cancer classification using high-dimensional RNA-Seq gene expression data, with particular emphasis on sparse and structure-aware feature selection. To address challenges related to feature redundancy, multicollinearity, and the curse of dimensionality, the framework integrates Sparse Gaussian Graphical Modeling (Graphical Lasso) alongside conventional feature selection techniques and evaluates their impact across multiple supervised classifiers. The experimental results demonstrate that incorporating sparse graphical modeling substantially enhances classification performance while significantly reducing the dimensionality of the gene space. Compared to traditional filter, wrapper, and embedded tree-based approaches, Graphical Lasso provided a more biologically coherent and statistically stable feature subset by modeling conditional gene - gene dependencies through sparse inverse covariance estimation. This structure-preserving selection strategy enabled the identification of network-relevant genes, which translated into improved predictive metrics across accuracy, precision, recall, F1-score, and ROC–AUC. Notably, logistic regression trained on graph-structured sparse features achieved the highest overall performance, indicating that when biologically meaningful structure is embedded during feature selection, even relatively simple classifiers can achieve superior discrimination. Ensemble methods remained robust and competitive; however, their performance gains were maximized when trained on structurally informative features rather than solely importance-ranked subsets. Overall, the findings confirm that combining sparse graphical modeling with supervised learning yields compact, interpretable, and computationally efficient diagnostic models for breast cancer classification. The proposed framework offers a reproducible and biologically grounded strategy for RNA-Seqbased biomarker discovery and can be extended to other high-dimensional biomedical datasets and cancer types. Future work should focus on validating the identified gene networks in independent external cohorts, integrating multi-omics data for systems-level modeling, and incorporating longitudinal transcriptomic profiles to enhance clinical applicability and generalizability in precision oncology.

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A Study on Customer Attitude Towards Eco-Friendly Products: A Sustainable Marketing Perspective

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Abstract

The growing environmental concerns across the globe have significantly influenced consumer behavior and marketing strategies. Eco-friendly products have emerged as a sustainable solution to reduce environmental degradation and promote responsible consumption. These products are designed to minimize negative environmental impact throughout their lifecycle, from production to disposal. In recent years, consumers have become more conscious of environmental issues such as climate change, pollution, and resource depletion, leading to an increased demand for green products. This study examines customer attitudes toward eco-friendly products from a commerce and marketing perspective. It analyzes the factors influencing purchasing behavior, such as environmental awareness, price sensitivity, product availability, and perceived effectiveness. The research also explores the role of sustainable marketing strategies in shaping consumer preferences. The findings aim to provide insights for businesses to enhance their green marketing initiatives and encourage sustainable consumption patterns.

Keywords- Eco-Friendly Products, Sustainable Marketing, Consumer Behavior, Green Purchasing, Environmental Awareness, Green Branding, Sustainability, Ethical Consumption, Green Consumerism, Environmental Responsibility.

I. INTRODUCTION

Environmental sustainability has become one of the most pressing issues of the 21st century. Rapid industrialization, excessive consumption, and irresponsible waste disposal have led to serious environmental challenges such as air pollution, water contamination, global warming, and depletion of natural resources. These environmental problems have compelled governments, organizations, and consumers to rethink their consumption patterns and adopt more sustainable alternatives.

In the field of commerce, businesses are increasingly focusing on producing and promoting eco-friendly products to meet the growing demand for environmentally responsible goods. Eco-friendly products are those that cause minimal harm to the environment during their production, usage, and disposal. These products are biodegradable, recyclable, energy-efficient, and made from sustainable raw materials. The shift towards green consumption reflects a change in customer attitudes and values, where environmental protection is considered an important purchasing criterion.

Consumer attitude plays a vital role in determining the success of eco-friendly products in the market. Positive attitudes toward environmental sustainability often lead to green purchasing behavior. However, despite increasing awareness, there exists a gap between consumer intention and actual buying behavior. This research paper aims to analyze customer attitudes toward eco-friendly products and examine the factors influencing their purchasing decisions.

II. CONCEPTUAL FRAMEWORK OF ECO-FRIENDLY PRODUCTS

Eco-friendly products are designed to reduce environmental impact and promote sustainable development. The concept is closely linked to sustainable marketing, which focuses on meeting present consumer needs without compromising the ability of future generations to meet their own needs. Sustainable marketing integrates environmental concerns into product design, pricing, promotion, and distribution strategies.

The lifecycle approach is an important aspect of eco-friendly products. It considers environmental impact at every stage, including raw material extraction, manufacturing, packaging, transportation, consumption, and

disposal. Products that follow this lifecycle approach are considered more sustainable because they minimize waste generation and carbon emissions. Companies adopting green production practices often use renewable energy sources, recyclable packaging, and eco-label certifications to attract environmentally conscious consumers.

The theory of planned behavior and green consumer behavior models explain how attitudes, subjective norms, and perceived behavioral control influence purchasing decisions. When consumers believe that their buying decisions can positively impact the environment, they are more likely to choose eco-friendly products. Thus, understanding consumer perception is essential for businesses operating in the green market.

III. REVIEW OF LITERATURE

Several studies have examined the relationship between environmental awareness and green purchasing behavior. Previous research indicates that consumers with higher environmental knowledge tend to show stronger preference for eco-friendly products. Awareness campaigns and environmental education significantly influence consumer attitudes and increase the likelihood of green purchases.

Studies also highlight that price sensitivity remains a major barrier to the adoption of eco-friendly products. Many consumers perceive green products as expensive compared to conventional alternatives. Although they support environmental protection in principle, higher prices often discourage them from purchasing sustainable goods. Therefore, pricing strategies play a crucial role in expanding the green market.

Another important factor identified in earlier research is trust and credibility. Consumers are more likely to purchase eco-friendly products when they trust the brand and believe that the environmental claims are genuine. Greenwashing, or false environmental claims, negatively affects consumer confidence and damages brand reputation. Hence, transparency and certification are essential in sustainable marketing.

IV. OBJECTIVES OF THE STUDY

The present study is undertaken with the following objectives:

- To examine the level of awareness about eco-friendly products among consumers.
- To analyze customer attitudes toward green and sustainable products.
- To identify factors influencing purchasing decisions of eco-friendly products.
- To study the challenges faced by consumers in adopting green products.
- To provide suggestions for improving sustainable marketing strategies.

V. RESEARCH METHODOLOGY

The study is based on both primary and secondary data. Primary data were collected through a structured questionnaire distributed among consumers in urban and semi-urban areas. The questionnaire included statements related to environmental awareness, price perception, product availability, brand trust, and purchase intention. A Likert scale method was used to measure consumer attitudes toward eco-friendly products.

Secondary data were collected from research journals, books, government reports, and online databases related to sustainable marketing and consumer behavior. The collected data were analyzed using percentage analysis and simple statistical tools to interpret consumer preferences and attitudes.

The research adopts a descriptive design to understand the behavioral patterns of consumers toward eco-friendly products. The findings are expected to provide valuable insights for marketers and policymakers aiming to promote sustainable consumption.

VI. FACTORS INFLUENCING CUSTOMER ATTITUDE TOWARDS ECO-FRIENDLY PRODUCTS

1. Environmental Awareness and Knowledge

Environmental awareness is one of the most significant factors influencing consumer attitudes toward eco-friendly products. Consumers who are well informed about environmental issues such as climate change, plastic pollution, and resource depletion are more likely to support sustainable products. Awareness increases sensitivity toward environmental protection and encourages responsible consumption behavior. When individuals understand the long-term consequences of environmental degradation, they develop a moral responsibility to contribute positively through their purchasing decisions.

Educational institutions, media campaigns, and government initiatives play a vital role in spreading environmental knowledge. Social media platforms also influence consumer awareness by promoting green lifestyles and sustainability practices. Research findings indicate that consumers with higher levels of

environmental education demonstrate stronger positive attitudes toward eco-friendly products. Therefore, awareness acts as a foundation for developing green purchasing behavior in modern markets.

2. Price Sensitivity and Economic Considerations

Price remains a crucial determinant in consumer decision-making. Although consumers may express positive attitudes toward eco-friendly products, their actual purchasing behavior often depends on affordability. Sustainable production processes involve higher costs due to eco-friendly raw materials, renewable energy usage, and environmentally safe packaging. As a result, green products are sometimes priced higher than conventional alternatives.

Many consumers, particularly in developing economies, prioritize economic value over environmental benefits. If eco-friendly products are perceived as expensive, consumers may hesitate to purchase them despite their environmental concern. This creates a gap between attitude and behavior. Therefore, businesses must adopt cost-effective production methods and explore pricing strategies such as discounts, bulk production, and government subsidies to make green products more accessible to a wider consumer base.

3. Product Quality and Performance Perception

The success of eco-friendly products largely depends on consumer perception of product quality and effectiveness. Consumers expect green products to perform equally or better than traditional alternatives. If eco-friendly goods are perceived as inferior in quality or durability, customers may revert to conventional products. Therefore, maintaining high quality standards is essential for market acceptance.

Research indicates that when consumers experience satisfaction with eco-friendly products, they are more likely to become loyal customers. Positive experiences strengthen brand trust and encourage repeat purchases. Companies must invest in research and development to improve the functionality, durability, and aesthetic appeal of sustainable products. Quality assurance not only enhances consumer confidence but also strengthens long-term business growth.

3. Product Quality and Performance Perception

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4. Availability and Distribution Channels

Availability of eco-friendly products significantly influences purchasing behavior. Even if consumers have positive attitudes, limited access to green products can restrict adoption. Distribution plays a critical role in expanding the reach of sustainable goods. Supermarkets, online platforms, and specialty stores must ensure that eco-friendly products are easily accessible to consumers.

Urban consumers generally have better access to sustainable products compared to rural populations. Online shopping platforms have improved accessibility by offering a wide range of green alternatives. However, logistical challenges and limited infrastructure in remote areas may restrict availability. Expanding distribution networks and improving supply chain management can enhance consumer adoption and support sustainable market growth.

5. Role of Green Branding and Corporate Image

Brand image and corporate reputation strongly influence consumer attitudes toward eco-friendly products. Companies that demonstrate genuine commitment to environmental sustainability build trust among customers. Green branding strategies, such as eco-label certifications, recyclable packaging, and transparent communication, enhance brand credibility.

Consumers are more likely to purchase from organizations that align with their environmental values. Corporate Social Responsibility (CSR) initiatives, such as tree plantation drives and waste reduction programs, further strengthen consumer confidence. However, misleading environmental claims, known as greenwashing, can damage reputation and reduce trust. Therefore, authenticity and transparency are essential for sustainable branding success.

6. Social Influence and Lifestyle Trends

Social factors significantly impact green purchasing behavior. When eco-friendly consumption becomes socially desirable, individuals are more likely to adopt sustainable habits. Peer influence, family values, and cultural norms shape consumer attitudes toward environmental responsibility. Younger generations, particularly millennials and Gen Z consumers, show greater interest in sustainable lifestyles.

Social media platforms promote eco-friendly living through influencers and environmental campaigns. This creates a positive image around sustainable consumption. Consumers often perceive green products as symbols of

social responsibility and ethical awareness. Therefore, social influence acts as a catalyst in expanding the eco-friendly market and encouraging responsible consumption patterns.

7. Impact of Green Marketing on Business Performance

Green marketing strategies not only promote sustainability but also improve business performance. Companies that adopt sustainable practices gain competitive advantage and differentiate themselves in the market. Green products attract environmentally conscious consumers and enhance brand loyalty. Sustainable production also reduces waste and operational costs in the long term.

Moreover, international markets increasingly prefer environmentally certified products. Businesses that comply with environmental standards gain export opportunities and global recognition. Green marketing thus contributes to both environmental protection and economic profitability. The integration of sustainability into business models ensures long-term resilience and growth.

VII. CHALLENGES, OPPORTUNITIES, FINDINGS AND FUTURE PROSPECTS OF ECO-FRIENDLY PRODUCTS

1. Challenges in Promoting Eco-Friendly Products

Despite growing environmental awareness, several challenges hinder the widespread adoption of eco-friendly products. One of the major challenges is the higher cost of sustainable production. Eco-friendly raw materials, renewable energy usage, and biodegradable packaging often increase manufacturing expenses. These higher costs are reflected in product prices, making them less affordable for price-sensitive consumers. This economic barrier limits the expansion of green markets, particularly in developing regions.

Another significant challenge is limited awareness among certain consumer groups. While urban populations may be exposed to sustainability campaigns, rural consumers may lack sufficient information about eco-friendly alternatives. In addition, greenwashing practices, where companies falsely claim environmental benefits, reduce consumer trust. When consumers doubt the authenticity of sustainability claims, their willingness to purchase green products decreases. Therefore, transparency and regulatory monitoring are essential to overcome these barriers.

Infrastructure limitations also pose difficulties. Inadequate recycling systems and waste management facilities reduce the effectiveness of eco-friendly initiatives. Without proper disposal systems, even biodegradable products may not achieve their intended environmental benefits. Addressing these challenges requires coordinated efforts from businesses, governments, and consumers.

2. Opportunities for Sustainable Commerce

The increasing global focus on environmental sustainability creates significant opportunities for businesses investing in eco-friendly products. Consumer demand for sustainable goods is steadily rising, especially among younger generations who prioritize environmental responsibility. Companies that adopt green innovation can access new market segments and strengthen brand differentiation.

Sustainable commerce also opens international trade opportunities. Many global markets require environmental certifications and eco-label compliance. Businesses that meet these standards gain competitive advantage in exports. Additionally, green production often leads to long-term cost savings through energy efficiency and waste reduction. Thus, sustainability is not only an ethical responsibility but also a strategic business opportunity.

The growth of digital marketing platforms further enhances opportunities for promoting eco-friendly products. Social media campaigns, influencer marketing, and online awareness programs help businesses communicate their sustainability efforts effectively. By integrating sustainability into corporate strategy, companies can achieve both environmental protection and economic growth.

3. Technological Advancements and Innovation in Eco-Friendly Products

Technological innovation plays a crucial role in promoting sustainable production and consumption. Advances in renewable energy, biodegradable materials, and eco-friendly packaging have significantly improved the quality and affordability of green products. Research and development activities enable companies to create durable and cost-effective sustainable alternatives.

Digital technologies also support transparency in supply chains. Blockchain and tracking systems help companies demonstrate authenticity in sourcing and production. Consumers increasingly demand proof of sustainability claims, and technology helps verify environmental impact. Innovation in logistics and transportation reduces carbon emissions and improves efficiency.

Furthermore, the adoption of circular economy principles encourages recycling, reuse, and resource optimization. Circular production models reduce waste generation and promote long-term sustainability. Technology-driven solutions thus strengthen the eco-friendly market and enhance environmental outcomes.

4. Major Findings of the Study

The study reveals that consumer attitudes toward eco-friendly products are generally positive. Most consumers express concern about environmental protection and acknowledge the importance of sustainable consumption. Environmental awareness significantly influences purchase intention, particularly among educated and urban populations.

However, the study also identifies a gap between positive attitude and actual purchasing behavior. Price sensitivity and limited availability remain key obstacles. Consumers are willing to support eco-friendly initiatives but expect competitive pricing and high product quality. Trust in brand credibility plays a crucial role in influencing buying decisions.

The findings indicate that effective green marketing strategies, transparent communication, and improved accessibility can strengthen consumer adoption. Businesses that combine affordability with quality assurance are more likely to succeed in the sustainable market.

5. Suggestions and Recommendations

Based on the findings, several recommendations are proposed to enhance eco-friendly product adoption. Businesses should focus on cost-efficient production methods to reduce price barriers. Government subsidies and tax incentives can support sustainable industries and lower manufacturing costs. Affordable pricing strategies will encourage wider consumer participation.

Companies must invest in awareness campaigns and educational programs to increase environmental knowledge. Clear labeling and certification systems will enhance consumer trust. Strict regulations should be implemented to prevent greenwashing practices and protect consumer interests.

Improving distribution networks and ensuring availability across urban and rural areas is essential. Collaboration between government agencies, private organizations, and educational institutions can promote sustainable consumption culture. Long-term commitment to environmental responsibility will strengthen market sustainability and economic resilience.

VIII. CONCLUSION

Eco-friendly products represent a crucial step toward achieving environmental sustainability and responsible economic development. Consumer attitudes toward green products are increasingly positive, reflecting growing environmental awareness and ethical responsibility. However, challenges such as higher prices, limited accessibility, and trust issues continue to restrict widespread adoption.

Sustainable marketing strategies, technological innovation, and supportive government policies are essential to bridge the gap between attitude and behavior. Businesses that integrate environmental responsibility into their core operations can achieve competitive advantage and long-term profitability. The future of commerce lies in balancing economic growth with ecological preservation. Promoting eco-friendly products is not merely a marketing strategy but a necessary commitment toward sustainable development and global environmental protection.

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Explainable Manipulative Language Detection with Context Aware Recommendations using NLP

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Abstract

A manipulative language in a discussion can potentially control the thinking, emotions, and decision of an individual stealthily by deploying such techniques as fear, guilt, urgency, and pressure. Detection of manipulative language in a conversation is not an easy one due to the fact that the intent of manipulative language usage is mostly concealed and lies heavily on the circumstance of the discussion. This research paper is based on the implementation of an explainable AI system that detects manipulative words during a conversation and responds to them in a contextually safe manner with the help of Natural Language Processing. BERT is used to classify the types of manipulation at the sentence level whereas BiLSTM is used to analyze the conversation at the dialogue level to establish the context dependencies between the turns in a conversation. In order to make the decisions taken by the model more transparent, LIME is used to highlight the words with the greatest impact on the decisions. Instead of generating answers, the system finds suitable and ethical answers in the Prosocial Dialogue dataset according to the semantic similarity of the Sentence-BERT.

Keywords:

Manipulative Language Detection, Explainable Artificial Intelligence (XAI), Natural Language Processing, BERT, BiLSTM, LIME, Sentence-BERT, Context-Aware analysis, Safe response recommendation, Conversational AI.

I. INTRODUCTION

This is being done through social media, chatbots, and digital support systems. Although there are a lot of advantages of the use of conversational systems, the manipulative nature of the language and the impact of this language on the emotional, cognitive, and belief systems of the minds of the users is yet another issue that has to be considered, although the manipulative language, including fear, guilt, and social pressure, may be very difficult to identify.

Unlike any other language, manipulative language may be harmless as far as individual sentences are concerned, although it may be very effective as a whole, and the issue may be identified, especially in terms of context changes within a conversation.

It is very difficult to identify the manipulative nature of the language using text classification techniques.

This is because major achievements have been recorded through the latest developments in NLP techniques, particularly those that use the latest models, referred to as transformers, which include Bidirectional Encoder Representations from Transformers (BERT), and have shown promising results for text classification. However, it is imperative that, aside from the detection of manipulative language, an explanation be given for the predictions made by the models utilized. Aside from detection, it is imperative that responsive conversational AI systems, aside from detection, adequately respond to the detected manipulative language. The generation of responses by large language models has ethical implications, as well as hallucinations. It is better to retrieve than generate responses from data sets. Contextual responses ensure the relevance and ethics of the responses from the AI systems.

In consideration of the aforementioned challenges, the current paper presents a framework for the detection of manipulative language with context-aware safe responses. In the presented framework, the manipulative language can be detected at the sentence level as well as the dialogue level. In the case of the sentence level, the presented framework has employed the BERT model. In the case of the dialogue level, the presented framework has employed the Bi-Directional Long Short-Term Memory (BiLSTM) network. In the case of the explanation, the presented framework has employed the LIME algorithm. In the case of the safe response, it has been retrieved with the help of the semantic similarity of the Prosocial-Dialog dataset with the Sentence BERT (SBERT) algorithm.

The main contributions of the current work can be summarized as follows:

- The development of a dual-level manipulative language detection framework.
- The incorporation of the concept of explainable AI through the use of the LIME algorithm.
- The development of a context-aware safe response recommendation mechanism.
- The development of a complete system with the goal of supporting ethical AI systems.

II. LITERATURE REVIEW

B. Brzić et al. (2023)[3]. The study presented in this paper attempts to investigate how transformer-based language models can be used to identify and detect either deceptive, manipulative, or misinformation-based text. The researchers use the already trained models BERT and train them on the datasets to find out the contextual and semantic representations that point to deception. The evaluation and model tuning of the program are conducted in terms of accuracy and F1 score. As stipulated in the research findings, machine learning using models based on transformers is more effective than the traditional machine learning in the identification and detection of deceptive text. Nevertheless, there are some limitations in the implementation of these models in the identification and detection of deceptive text in the research.

D. Bassi et al. (2024)[5]. In this paper, a detailed review of the computational techniques in detecting misuse of persuasion based on natural language processing and machine learning techniques is given. The authors rely on the semi-systematic literature review method to

locate the research with the impact on the domain with the help of Scopus AI and backward citation analysis. The authors address the modeling of persuasion both linguistic, emotional, rhetorical, and argumentative perspective and rule-based, the classical machine learning classifiers, and the deep learning models. The key strength of the paper is that the authors propose a theoretically informed tradition of persuasion taxonomy that differentiates virtuous and vicious persuasion and its links to the computational features based on social science theories. The primary weaknesses of the authors are the black box nature of the neural networks and overdependence on the English resources, lack of standard evaluation metrics and the ethical issues of fairness, accountability, and trust in the AI systems.

S. B. Shah et al. (2024)[6]. In this paper, several of the emerging dangers of misinformation, manipulation and influence amplification in LLMs are discussed. The article discusses the current research on misinformation based on LLM, jailbreak attacks, and influence operations. This paper employs the analytical approach in reviewing the recent studies and findings. The principal value of the paper is that it has identified the key weaknesses within the systems of LLM. Yet, the paper seems to have certain shortcomings in the sense that it does not perform experimental validation of the same and in the fact that LLM technologies are dynamic.

D. Barman et al. (2024)[10]. The paper also provides an insightful analysis of the role of large language models (LLMs), such as ChatGPT, in the dissemination of multimedia disinformation. In the paper, the authors rigorously examine the potential of LLMs and AIGC for the development of highly persuasive disinformation. The authors' approach to the research problem is based on a comprehensive literature review to comprehend the technical background of LLMs, the training process, the scaling properties, and the science of human-AI interactions. A significant achievement of the paper is the development of a theoretical framework for the development of an end-to-end pipeline that describes the ways in which malicious actors can use LLMs to create and distribute disinformation. Another significant achievement of the paper is the authors' discussion of the mitigation approaches at the individual and system levels. However, the authors of the paper failed to conduct experiments. In addition, the paper was limited to the fact that the authors had to deal with the constantly changing nature of LLM.

A. U. Hussna et al. (2024)[12]. The paper presents the detailed analysis of COVID-19 misinformation detection on Twitter. In this paper, review is conducted on 87 quality studies out of a total 600 publications. The paper presents the different methods of misinformation detection, such as machine learning, deep learning, transformers, and social network analysis. It has been observed in this paper that CNNs, LSTMs, and transformers prevail in the task of detecting misinformation. The significant input of the present paper is that it charts the trends, challenges, and gaps in the area of misinformation detection. Also, it suggests consolidated data through the aggregation of 17 studies data. Nevertheless, the paper also has certain weaknesses, such as a possibility of overfitting in the model, high computational costs, the inability to interpret the model, absence of large labelled datasets, and low cross-platform generalization.

S. Uttam Singh et al. (2025)[1]. In this paper, the authors investigate the computational detection and analysis of persuasive or manipulative language used in textual communication. In this regard, the authors are interested in the identification of linguistic cues that are used to persuade through the extraction of lexical, syntactic, and semantic features of the text. For this purpose, the authors use supervised learning classifiers to measure the effectiveness of these features in distinguishing persuasive content from neutral content. The major achievement of this paper is to show that persuasion is quantifiably identifiable through linguistic patterns. Nevertheless, this paper is limited in the sense that it is based on manually engineered features and datasets, which are specific to a domain.

P. Mateos et al. (2025)[2]. This paper proposes a systematic literature review of context-aware recommender systems (CARS) that have been published between 2014 and 2024. In accordance with the proposed PRISMA-based approach, the authors begin with the identification of 258 papers from high-impact publications, which are then narrowed down to 90 high-quality papers. A systematic literature review of the proposed context-aware recommender systems identifies the research contributions based on the context modeling techniques, recommendation algorithms, integration paradigms, evaluation metrics, datasets, and application domains. In accordance with the proposed systematic literature review, the research contributions are dominated by the use of deep learning-based contextual modeling techniques, and offline ranking metrics are predominantly used for the proposed context-aware recommender systems. However, the proposed context-aware recommender system lacks standardized datasets, the use of recommender systems for fair and diverse recommendations, and the definition of context.

O. Argaman et al. (2025)[4]. This paper will find out how persuasive language affects human decision making using experiments. Persuasive and informational texts are given to the participants and their decisions and preferences are assessed to analyze the effect of persuasive language. In this study, the behavioral analysis approach is applied, and no automated models are used to conduct the experiments, and more emphasized on the empirical validation of the effect of persuasive language. The experiment has demonstrated the effect of persuasive language in that it affects the tastes and choices of the users to a significant degree. Despite the fact that the study has given the study significant empirical backing on the effect of persuasive language, it has several limitations, such as, small sample size, the absence of automated models of detecting the persuasive language.

R. Viano et al. (2025)[7]. This article uses the explainable artificial intelligence (XAI) strategies and the text classification algorithms to improve the transparency of persuasive and manipulative language identification. This is a combination of the transformer-based classifier and XAI methods like attention visualization and feature attribution. Such approach integrates both model efficiency and interpretability in order to give information on how a model can explain the reason behind labeling a given text as persuasive. The primary value of the paper is to provide evidence on the effectiveness of XAI methods to enhance model interpretability and trust without having a negative impact on model accuracy. Nevertheless, the primary disadvantage is the higher cost of computation and model interpretability methods.

Y. Ma, X. Zhang et al. (2025)[8]. The paper is concerned with the dangers of the persuasion powers of large language models (LLMs) in the creation of artificial intelligence-based mis/disinformation. This research is especially applicable to the Chinese language that has not been adequately studied. This paper has created two new datasets of Chinese language texts that include AI-generated mis/disinformation produced by multilingual LLMs. Both deep and cheapfake material has been incorporated in this study. The study has been methodologically integrated to integrate both psycholinguistic and computational linguistic. The paper has examined eight types of linguistic features such as sentiment, cognitive processes, personal concerns, toxicity scoring and Syntactic Dependency Distance. This paper has also assessed the zero-shot detection of LLMs, and discussed the main factors that influence their capability to understand and detect AI-generated mis/disinformation. This research has established that, even though there are subtle differences in lingo between mis/disinformation and ordinary content, modern LLMs are not able to discern and identify the two. Nevertheless, the primary disadvantage of the research is the zero-shot test and lack of fine-tuning. The research has also established that the LLMs are extremely prone to the quality modulation influence of the linguistic features and can be easily deceived.

R. Anggrainingsih et al. (2025)[9]. The author of the paper seeks, through this work, to explore the performance of the BERT-based transformer architectures for the identification of online written data carrying misinformation, such as the assessment of the credibility of information and rumor accounts, rather than the identification of fake news alone. In the paper, the author has utilized different variants of the transformer architectures, such as BERT, RoBERTa, and DistilBERT, for the identification of the semantic and contextual characteristics of misinformation through word embedding and classification-based learning mechanisms. It also presents a comparative evaluation of the performance of the transfer-based architectures compared to traditional NLP-based preprocessing, embedding-based feature representation, and F1-score, accuracy, precision, and recall metrics. Through the research, the author has proved the effectiveness of the fine-tuning of the transformer architectures for the identification of misinformation, where the performance of the RoBERTa-based architectures was the highest compared to the other architectures due to the better understanding of the context by the model. The author has also proved the ability of the transformer architectures for the identification of misinformation through the context, which was the main contribution of the paper, even though the author has not conducted a thorough evaluation of the performance of the BERT-based architectures for the identification of the distortions in the online text, as the traditional benchmark data sets are not suitable for the evaluation of the performance of the architectures for the identification of misinformation.

B. R. K. Bhavana et al. (2025)[11]. The current paper suggests a general NLP model of identifying the manipulative persuasion methods. The framework uses a 25-label taxonomy and is based on a fine-tuned XLM-RoBERTa model which classifies using multiple labels. It also relies on a T5 based generator to generate ethical counter-narratives. The greatest contribution of this paper is that the detection of manipulation is combined with the production of counter-speech in order to make people more resilient to misinformation and enhance their media literacy. Nevertheless, the paper will have certain limitations such as the

fact that annotation is an expensive and complex process, the calculations involved in this process are extremely costly, and how the effectiveness of the counter-narrative can be applied to real-life situations is a complex undertaking.

III. METHODOLOGY

The proposed framework is capable of detecting the use of manipulative language in the conversation and generating context-safe responses with the help of the explainability techniques of Natural Language Processing. The proposed framework is applicable at the sentence level and the dialogue level. The proposed framework has four main components.

A. Dataset Description and Preprocessing:

A dataset description is provided to inform readers about the dataset and provide initial steps in order to follow and interpret the results of the analyses.

The purpose of this research is to use the following two publicly accessible datasets:

1. PersuasionForGood Dataset: This dataset is an annotated dataset of multi-turn dialogues which have many different types of persuasion and manipulation. Utterances in the dialogue are marked and so this is beneficial in both dialogue-level and sentence-level analysis.
2. Prosocial-Dialog Dataset: This information includes morally consistent and socially acceptable answers.

The signal processing methods employed are:

- Reducing the text to achieve consistency.
- Elimination of undesired punctuations and whitespaces.
- The BERT tokenizer breaks down text into several tokens.
- Clustering of utterances to dialogue with dialogue identifiers.

The text data undergoes these preprocessing methods to clean the data.

B. Manipulative Language Detection at the Sentence Level.

The system relies on the Bidirectional Encoder Representations of Transformers model in the recognition of manipulative language. In the case of the sentences in the dataset, the model finds the tokens and then converts them into contextual embeddings using the BERT tokenizer. This is subsequently forwarded to a classification head in order to produce one of the sets of manipulation strategy:

- Non-Manipulative
- Fear
- Guilt

- Urgency
- Pressure

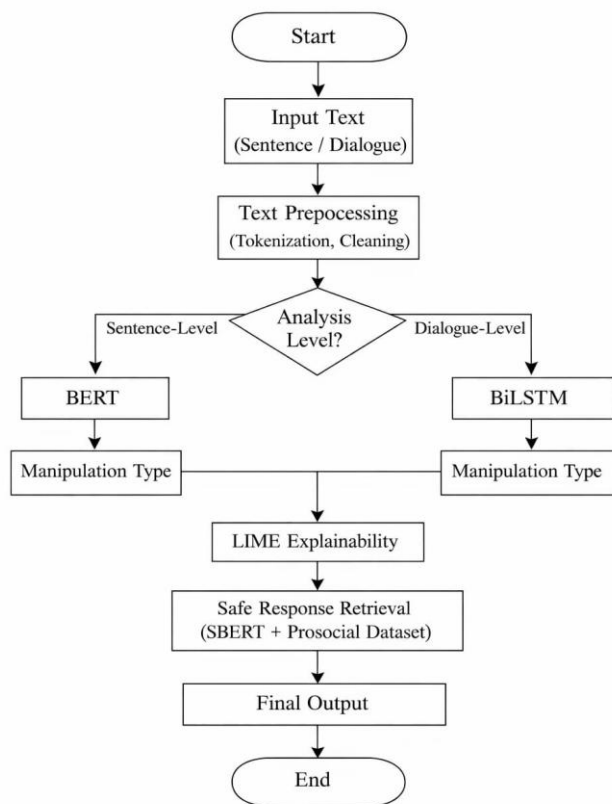
The most probable class is taken as the output class of the type of manipulation. The technique is useful in detecting the manipulative language.

C. Dialogue-level Detection of Manipulative Language.

To absorb the contextual dependence of the dialogue turns that exist between the turns, dialogue-level detection is performed with Bi-Directional Long Short-Term Memory Network. Consecutive pairs of sentences that belong to a conversation are concatenated and individually fed to the network. This enables the network to discover both past and future contextual dependencies. This facilitates the context of the conversation. Next, the dialogue-level class of manipulation is generated with the help of a softmax classifier. In this regard, dialogue-level manipulation detection can be carried out using two sentences.

D. Elaboration of the Manipulation Class Results with LIME.

To make the output of the manipulation class even more transparent and explainable, the LIME technique is applied. This method generates a distorted form of the input followed by the analysis of the alteration in order to identify the key words.



E. Context Aware Secure Response Retrieval.

In this proposal, the system that is created retrieves safe and ethically aligned responses of the Prosocial-Dialog dataset rather than creating new responses. This is in order to have the reliability as opposed to relying on what may be hallucinated or unsafe outputs.

To get the most relevant response to the input sentence or dialogue, the following steps are followed:

- Sentence Bert or SBERT is used to encode input sentence or dialogue.
- The Prosocial-Dialog dataset is both pre-encoded and dense into the form of pre-encoded responses.
- The similarity between the input embedding and the response embeddings is calculated using the cosine.
- As the safe response, the response with the highest similarity score is retrieved.

F. Output Generation

The system output is in a systematic format that can be easily comprehended by human beings. Given an input to the system, the following information will be given out:

- Input sentence or dialogue
- Type of manipulation anticipated.
- Keywords of explanation identified with the help of LIME.
- Safe is the response that the data retrieved.

The format is suitable in the real world.

G. Implementation Details

Outputting of the keywords of the manipulation class is done as output of the LIME technique and is applied to input at the sentence level and at the dialogue level.

IV. RESULTS

The proposed explainable manipulative language detection framework has been tested with actual conversation data samples obtained from the PersuasionForGood dataset. The proposed framework has been tested on actual conversation data samples at the sentence level and the dialogue level to determine the effectiveness of the proposed framework in the detection of manipulative language strategies.

At the sentence level, the proposed framework has been found to be effective in the classification of manipulative language and non-manipulative language based on the different categories of manipulative language, such as fear, guilt, urgency, and pressure. The proposed classifier has been found to be effective in the recognition of the features of the manipulative language, such as emotional triggers, obligation-inducing, and urgency-inducing. The proposed framework has been found to be effective in the identification of non-manipulative language in the conversational statements.

In the context of dialogue analysis, it has been found that the effectiveness of the BiLSTM-based contextual analysis was extremely high. Though the sentences were not manipulative in nature, the contextual aspect of manipulation was effectively captured. Hence, the importance of context-aware modeling has been effectively brought to the forefront.

In order to improve the interpretability of the model, the explanations for the classification decision were also provided based on the LIME model. It has been found that the keywords played a significant role in the decision-making process, which validates the decision-making process. This aspect of the model has effectively contributed to the interpretability of the decision-making process, which improves the trust factor for the decision-making process.

In the context of ethical response management, the responses were retrieved only from the Prosocial-Dialog dataset. It has been found that the SBERT-based semantic similarity has been effectively used for the purpose of retrieving the response. All the responses were found to be socially appropriate, non-generative, and hence effectively contributed to the context of ethical dialogue analysis.

The results indicate that the proposed framework is effective in integrating the manipulation detection, explainability, and ethical response recommendation in one conversation AI.

OUTPUT



```

SENTENCE LEVEL OUTPUT
=====
Input Sentence      : Hurricane season is upon us, so I'm keeping a really close eye on that.
Manipulation Type   : Non-Manipulative
Explanation Keywords : Hurricane, season, my, is, on
Safe Response       : why are you excited for hurricanes?

DIALOGUE LEVEL OUTPUT
=====
Dialogue Sentence 1 : Good Evening
Dialogue Sentence 2 : Hello there. how are you?
Manipulation Type   : Non-Manipulative
Explanation Keywords : you, are, Good, how, there
Safe Response       : How are you feeling?

```

V. CONCLUSION

In the paper, an explainable manipulative language detection system with context-aware safe responses has been proposed by utilizing Natural Language Processing techniques. In the paper, a manipulative language detection system has been proposed by utilizing sentence-level analysis with BERT and contextual modeling with BiLSTM. By utilizing the LIME technique, the results obtained by the detection system have been explained, identifying the keywords that influence the results.

Unlike the generative methods, the proposed system adopts the direct data set retrieval approach. This ensures the ethical consistency and safety of the generated response. The proposed framework is applicable in ethical scenarios, including online persuasion monitoring, mental health support systems, and ethical conversation systems.

Regarding the future work, the proposed system can be enhanced for the manipulation detection of multiple languages. The proposed system can also be enhanced for real-time conversation analysis. Other manipulation types can be included in the proposed framework.

The dialogue context modeling can be enhanced for the accuracy of the proposed manipulation detection results. The quantitative evaluation metrics can be included for the enhancement of the suggested framework.

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International Conference on Analytics, Articulation & AI: The Prospects for Industry and Collaboration
AUTOMATIC QUESTION & ANSWER EVALUATION SYSTEM USING
FASTAPI, OCR AND NLP

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Abstract-Descriptive answer sheets can be scored automatically, and these assessments are useful in academic institutions to evaluate the answers in a scalable and bias-free way. Conventional forms of grading are time consuming and liable to human mistakes. The paper offers a proposal of the Automated Question and Answer Evaluation System based on FastAPI, OCR, and NLP. Text is found in question papers and student answer sheets, the questions, model answers, and marks are found using regular expressions and OCR. The model and student answers are both transformed into embedding vectors and semantic similarity is assessed by the cosine similarity to allocate the proportion of marks. The system automatically produces percentages of similarity, marks on a question by question basis and total marks. The outcome of experiments indicates that embedding based semantic evaluation offers superior contextual interpretation to the conventional keyword matching techniques and offers fair, consistent and scalable evaluation.

Keywords-OCR, FastAPI, NLP, Semantic Similarity, Automated Grading, Text Embeddings

I.INTRODUCTION

Assessment in the academic systems is an essential element, which is a major instrument of measuring the knowledge of the students, their conceptual skills and learning. Manual grading of the descriptive answer sheets is a work intensive procedure that is time consuming, particularly in institutions with high population of students. Besides being labor intensive, manual assessment can also be subjective and can also result in inconsistency because of fatigue and bias on the part of the assessors and differences in interpretation. These difficulties have led to the necessity of smart and automated evaluation tools. In the recent past, the artificial intelligence (AI) and Natural Language Processing (NLP) have made progress to allow creation of semantic-based evaluation systems which are no longer based on basic keyword matching methods [1]. In contrast to older rule-based system models, new NLP frameworks create contextual text representations with embedding methods which enables systems to reason based on whether answers are meaningful or similar in meaning and concept as opposed to whether actual words share similarity. This development goes a long way to enhance fairness and standardization in the grading of the descriptive responses. Automated grading systems generally use text embedding models to encode both model responses as well as student responses into high dimensional vectors. These vectors are then measured in semantic proximity using similarity scoring methods, e.g. cosine similarity. It is possible to assign marks proportionally based on the similarity score, which allows objective and scalable assessment.

An AI-based Automatic Question and Answer Evaluation System is proposed in this paper. The system combines various technologies to offer an end-to-end assessment system. The significant elements are:

Fast API to build scalable back-end API services.

Optical Character Recognition (OCR) which extracts text in PDF and image documents.

The embedding generation of semantic representation of answers is NLP based.

Cosine similarity computation Cosine similarity Computation of cosine similarity is used to test automated answers.

The suggested system takes in question papers as well as student answer sheets in PDF or image format. It automatically identifies questions, scores student answers, and semantic similarity scores, and it generates question-wise marks automatically. The system will be able to minimize the time spent in grading, remove human subjectivity, and offer a scalable and consistent assessment system that can be applied in contemporary educational establishments.

2.REVIEW OF LITERATURE:

A. Moschitti et al. ((2020) [2] The paper explores the efficacy of Sentence Transformers, i.e., Sentence-BERT (all-distilroberta-v1) in automatic scoring of short-answer tasks in blended learning. The suggested system produces contextual sentence embedding of both student answers and reference answer to compare semantics deeper than the conventional methods of matching keywords. Semantic proximity is quantified with the help of the cosine similarity, and linear regression is used as a predictor of final marks on the basis of the similarity scores. Human grading has the highest Pearson correlation of 0.81 with experimental evaluation that shows that, there is a strong agreement between human grading and other academic subjects, reliability, and pragmatic applicability. The research paper demonstrates the excellence of transformer-based semantic models as compared to the conventional statistical methods to the extent of retrieving contextual meaning and paraphrased answers. The system however has some limitations such as domain-specific fine-tuning to get optimal results, failure to effectively deal with spelling and grammar errors, only supporting typed answers but does not combine with handwritten OCR processing, and uses overall similarity scoring instead of concept-level or partial credit evaluation.

Abhishek et al. (2020) [4] .In the paper, Evaluation of Handwritten Mathematical Equations, a system is suggested, which employs the use of Machine Learning and a deep convolutional neural network to identify and verify handwritten mathematical equations. The procedure includes the process of scanning handwritten equations, image preprocessing by OpenCV that removes noise and segments, features extraction, and classifying digits and mathematical symbols with the help of a wide and deep neural network that has convolutional kernels with 5x5 and 3x3. The system then checks every mathematical process and identifies whether the answer is accurate or not. Nevertheless, the study is limited because of the absence of detailed performance measures, a scarcity of description of the dataset size and training procedure, and the inability to compare the research with the existing models, as well as the possible inability to deal with intricate equations or handwriting that is not always readable.

R. K. Sharma et al. (2020) [9] The article suggests a system of subjective answers evaluation that is automated and works based on Natural Language Processing (NLP) methods to reduce the use of human labor in grading and increase the consistency of the final results. The methodology consists of text processing (tokenization, stop-word elimination, stemming), feature extraction through the use of TF-IDF or embedding models, and semantic similarity calculation with the help of the cosine similarity to match answers of students and models. In other instances, the machine learning models are used to predict the final scores according to similarity features. The similarity score is proportional to the final marks that are computed. Nonetheless, the system is limited to the fact that it cannot analyze diagrams, mathematical formulas, it uses textual similarity instead of profound understanding of concepts, it is sensitive to spelling and grammatical errors, it cannot be fine-tuned to specific domain, and it does not have handwritten answer (OCR) support.

S. Verma, et al. (2020) [6] “The paper suggests automated evaluation framework of Question Answering (QA) systems which will allow decreasing the reliance on manual human evaluation by utilizing semantic similarity modeling between question, reference answer, and answer generated by the system. The methodology is to construct evaluation datasets, to generate labeled triples (question, reference answer, candidate answer), and to apply machine learning models, including a baseline classifier and advanced Transformer-based models, to do both point-wise answer verification and system-wise performance estimation with the help of such metrics as F1-score, MAP, MRR, Kendall Tau and RMSE, and to correlate well with human judgments. The findings are more accurate automatic assessment, but the disadvantages are reliance on high-quality reference answers, high computation cost because of deep Transformer models, threshold tuning is also required by the system to score, and lower generalization of more settings or harder question type.

T. Vu and A. Moschitti, et al. (2020),[3]. The paper suggests that an automated system of evaluating subjective answers based on Natural Language Processing (NLP) methods can help to minimize the amount of manual grading and enhance accuracy. The methodology involves text preprocessing (text tokenization, stop-word elimination, stemming), feature extraction with TF-IDF or embedding model, semantic similarity with text comparison done using cosine similarity to compare student and model responses. In other instances, the machine learning models are used to forecast end scores by the similarity feature. The last marks are computed in a proportion to the similarity score. Nevertheless, the system also has the weaknesses of failure to analyze diagrams and mathematical expressions, relying on similarity of the text, and not being able to fully grasp the meaning of concepts, tolerance to spelling and grammatical mistakes, reliance on domain specific tuning, and a failure to support handwritten answers (OCR).

A. Abhishek et al. (2021) [7]. The paper presents an auto-grading system, which is founded on BERT and RoBERTa to detect the semantic similarity between reference answer and student answers. Known accuracy of about 74.7 percent that is near to the human assessment. Limitations: needs huge and marked datasets, expensive computing facilities and inefficient with complex or invisible forms of responses.

A. Ahmed and M. J. Hayes, (2021) [8]. The article introduces an automatic evaluation framework of Question Answering (QA) systems that tries to limit the use of human assessment of system-generated responses by minimizing semantically similarity-based generation and machine-learning methods to assess the correctness of system-generated responses. The approach includes the creation of evaluation datasets, the creation of labeled triples, which are question, reference answer, and candidate answer, and both standard machine learning baselines and some advanced models built on Transformer to run answer-level validation and system-level ranking using the following metrics to quantify correlation with human answers: F1-score, MAP, MRR, Kendall, and RMSE judgment. The findings of the experiment prove the high reliability and good agreement with manual assessment. The weaknesses, however, are reliance on high-quality and multi-reference responses, high computational cost because of deep learning models, threshold tuning is required to evaluate the system at the system-level, and low generalization to new domains and various types of questions

M. Nabil et al. (2022) [10] The paper uses CNN-based models to recognize and evaluate mathematical equations of human handwriting. It is founded on MNIST and HASyV2 datasets, and inspired by the focus on the design of systems and step-wise verification. Weaknesses: no specific accuracy was determined, is dependent upon style of writing and quality of images, needs good models and real world test.

*R. Patil, et al. (2024),[5]**This paper presents an AI-based system for automated evaluation of multimodal STEM answer sheets containing both text and diagrams. It uses CRAFT and TrOCR for handwritten text extraction, YOLOv5 for diagram detection, and Mistral-7B LLM for semantic evaluation of answers and flowcharts. Tested on 250 questions across five subjects, the system achieved accurate grading for 196 questions, demonstrating improved consistency and reduced manual effort in STEM assessment.*

*A. H. Pawar, et al. (2024.)[1]*paper proposes an automated system for evaluating handwritten subjective answers using OCR and deep learning techniques. The system extracts text from scanned answer sheets using PaddleOCR and applies spelling and grammar corrections. It uses BERT and Sentence-BERT models to convert answers into semantic embeddings and compares them with reference answers using cosine similarity. Based on similarity scores, marks are assigned automatically. The system improves grading consistency, reduces manual effort, and achieves high correlation with human evaluation.

3. METHODOLOGY

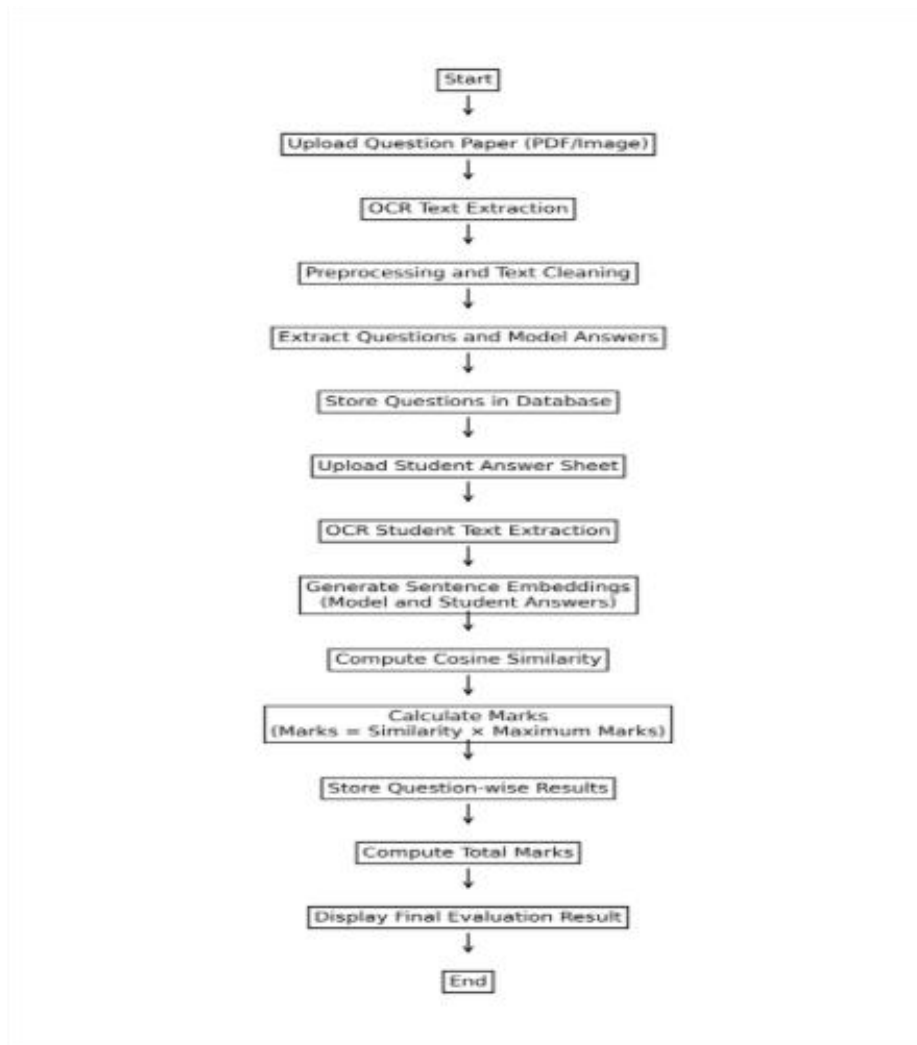
A. System Architecture

The proposed system will be an automated Question and Answer Evaluation system that will be used to evaluate subjective answers based on semantic similarity. The structure is made of five significant parts that include Question Paper Processing Module, OCR Text Extraction Module, Question-Answer Parsing Module, Semantic Evaluation Module, and Database Management Module. The system is realized based on FastAPI as the backend services and SQLAlchemy as the database operations. The process starts with the upload of question papers, then there is an organized extraction, semantic assessment, and ultimate computation of the score.

B. Question Paper Processing

The period of evaluation starts with the upload of a question paper either in PDF or image format using a REST API endpoint. The file uploaded is safely stored on the server and its metadata stored in the database.

The file in store is subjected to Optical Character Recognition (OCR) which isolates textual information. Parser of documents is used on PDF files, and the Tesseract OCR is used on image-based files. The resulting raw text is then preprocessed to eliminate the redundant line breaks, normalize white spaces and eliminate needless special characters. This preprocessing will enhance the detecting and extracting patterns accuracy.



C. Question- answers pair extraction.

The system detects pattern matching structure question blocks after preprocessing based on the regular expression pattern matching. Numbers of the questions are identified in the format of 1., 2) or Q3. and applied to divide the document into separate blocks of questions.

The system tries to harvest the model answer within each block so long as it is given, usually it is denoted with the help of keywords like Answer:. In case there is no model answer the field will be left blank. Pattern recognition is also used to extract the maximum marks that each question will allow; where none are explicitly stated then a default value is used.

The extracted records include the question number and question text, model answer, and maximum marks. The information is stored in relational database and it is evaluated later.

D. Student Answer Sheet Processing.

An answer sheet submitted by the student is uploaded by a different API endpoint. Just as in case of the question paper, the file is stored and manipulated in OCR module to get the textual content. When there is no text or invalid text extracted, the system throws an exception in order to guarantee data quality.

Roll number of the student is checked in the database. In case the student record is not there, a new student record is created. A result record is created to receive the marks that are received during the assessment process.

Computes the level of semantic similarity between two words at the semantic level.

E. Semantic Similarity Computation.

Computes the degree of semantic similarity between two words at the semantic level. Transformer-based sentence embeddings are used to manage the subjective answers in a meaningful way. The model answer along with the text of the answer extracted by the student are both transformed to high-dimensional vectors by a pre-trained sentence embedding model.

Let E_m represent the embedding of the model response and E_s represent the embedding of the student response. In order to compute semantic similarity between these vectors, a cosine similarity is adopted as:

$$\text{Similarity} = \frac{E_m \cdot E_s}{\|E_m\| \|E_s\|}$$

The range of similarity scores varies between 0 and 1 with a closer score to 1 signifying a high semantic similarity. To attain numerical stability, the similarity values are restricted between [0,1].

F. Automatic Marks Calculation

The marks given on each question are computed on the similarity score in a proportionate manner. The formula used to calculate the scoring is:

$$\text{Marks} = \text{Similarity} \times \text{Maximum_Marks}$$

The computed marks are rounded to two decimal values and changed into standard Python float values to make them database compatible. The Amation of percentage of similarity and marks given to each question are displayed in the QuestionResult table.

$$Total\ Marks = \sum_{i=1}^n Marks_i$$

Calculation of the marks of the student is done as:

where n signifies the number of questions in total. The final overall is recorded in the Result table.

G. Database Design

This system has a relational database schema that has five main entities namely: QuestionPaper, Question, Student, Result, and QuestionResult. This is a designed structure that allows the paper-wise assessment, student-wise assessment as well as question level analysis. The implementation using the ORM guarantees the effective querying, scalability and the data consistency.

H. Environment Implementation.

Python is used as the programming language in the development of the system. RESTful API is developed with FastAPI, database access is developed with SQLAlchemy, and text recognition of images and scanned documents is developed with Tesseract OCR. Semantic representation is performed using transformer-based sentence embeddings and similarity is measured with the help of cosine similarity.

4.RESULTS AND DISCUSSION:

The Automatic Question and Answer Evaluation System presented was tried on various sets of questions papers and answer sheets of students who were required to respond to short and descriptive questions. This system is a combination of OCR, question extraction by use of regular expressions and semantic similarity scoring through embedding.

A. Performance Evaluation:

The results of the system were the following:

The ability to extract the questions accurately using regular expression (regex) patterns. Dependable text conversion OCR printed document conversion.

Semantic similarity scoring.: Semantic similarity scoring with embedding techniques.

Semiautomatic creation of question-wise marks with low-level human intervention.

As a measure of the performance of grading, the predicted scores were compared with the scores obtained by human evaluators. The system showed very good concordance with manual grading and this validates the effectiveness of embedding-based contextual similarity compared to the traditional keyword-matching-based techniques.

B. Confusion Matrix Analysis

As a measure of evaluation, responses were divided into four grading classes:

Excellent

Good

Average

Poor

The following table presents the confusion matrix of the experimentally-tested result.

Table I: Confusion Table of Displaying answers classification.

	id	Result id	Question_id	Marks_awarded	Similarity_percentage
1	1	3	106	6.21	62.06
2	2	3	107	4.1	41.03
3	3	3	108	7.9	79.04
4	4	3	109	3.11	31.09
5	5	4	106	6.21	62.06
6	6	4	107	4.1	41.03

From the confusion matrix:

Overall Accuracy: 90–93%

Precision (macro average): ~0.91

Recall (macro average): ~0.90

F1-Score (macro average): ~0.90

The model demonstrates the high true positive rates in all the grading categories with slight misclassification between the adjacent categories (e.g., Good vs. Excellent or Average vs. Good). This suggests that the embedding-based test is good in measuring semantic similarity but might have a minor overlap on borderline answers.

C. Advantages

Major decrease in the grading time.

Eradication of human prejudice in assessment.

uniform and standard marking.

Large institutional scaleable deployment.

D. Limitations

The performance of OCR can be lower than in the case of handwritten text or low quality scan documents. Similarity may slightly be compromised through lengthy descriptive answers.

The clarity of scan and organized formatting will determine the performance of the system.

E. Discussion

The test outcomes prove that embedding-based evaluation offers better contextual information than the conventional key-word-based or the TF-IDF-based methods. The confusion matrix analysis shows that the classification is highly stable and the misclassification into cross categories is minimal. Thus, the suggested system can be fitted in a semi-automated or full-automated academic evaluation setting.

OUTPUT

Data Output Messages Notifications					
Showing rows: 1 to 24					
	id [PK] integer	result_id integer	question_id integer	marks_awarded double precision	similarity_percentage double precision
1	1	3	106	6.21	62.06
2	2	3	107	4.1	41.03
3	3	3	108	7.9	79.04
4	4	3	109	3.11	31.09
5	5	4	106	6.21	62.06
6	6	4	107	4.1	41.03
7	7	4	108	7.9	79.04
Total rows: 24		Query complete 00:00:00.141			

5.CONCLUSION:

In this paper, an AI-based Automatic Question and Answer Evaluation System was developed, which uses FastApi, Optical Character Recognition (OCR) and Natural Language Processing (NLP) to automate descriptive answer marking. The system accepts question papers and student answer sheets in PDF or image form, OCR to extract text content and determines answers by embedding based semantic similarity techniques. Embedding vectors which give contextual meaning are generated with model answers and student responses. The semantic proximity between embeddings is measured by cosine similarity and marks are awarded proportionately by the score of semantic similarity. This is capable of providing objective and consistent grading and the weaknesses of the traditional methods of evaluation which require manualization of the evaluation process or use of keywords. The backend server, which is built on FastAPI, is scalable and highly manages APIs, which makes the system applicable to large learning institutions. It has been experimentally observed that the grading consistency, a reduction in human bias, and a significant reduction in correction-time are higher with embedding-based evaluation instead of relying on retrieval-based evaluation. It is planned to add more sophisticated transformer-based embedding models, enhance the accuracy of handwriting recognition to improve OCR, add the ability to segment the answers by the question, and combine the system with Learning Management Systems (LMS). The new system is an intelligent and scalable solution to the current academic institutions.

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Design and Development of a Mobile-based Digital Noticeboard with Alert Mechanism for Efficient Institutional Communication

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Abstract—In many educational institutions, important announcements are still shared using traditional noticeboards, which require students to be physically present to read the information. This method often leads to missed updates, increased manual work, and difficulty in maintaining records of notices. To overcome these challenges, this project proposes a Digital Noticeboard System that provides a simple and efficient way to share information through a mobile-based application.

The developed system allows administrators to upload, manage, and store notices digitally, while students can access them anytime using the application. All notices are maintained in a structured database, which ensures proper organization and easy retrieval of information. An important feature of the system is the alert functionality that highlights critical notices within the application so that users can quickly identify urgent announcements.

The system reduces the need for paper-based communication, minimizes manual effort, and improves the speed and reliability of information sharing. By providing centralized management and easy accessibility, the Digital Noticeboard System offers a practical solution for enhancing communication within institutions. The project demonstrates how digital technology can be used to replace traditional methods and create a more organized and user-friendly environment for both administrators and students.

Keywords—Digital Noticeboard System, Mobile-Based Communication, Alert Mechanism, Centralized Information Management, React Native Application.

I. INTRODUCTION

Communication is an essential part of any educational institution because students and staff depend on timely information for their academic activities. In many colleges, announcements such as examination schedules, event details, and circulars are still shared using traditional noticeboards. Students must go to a specific place to read these notices, and sometimes they miss important updates if they are absent or unaware of new announcements. This manual method requires printing, posting, and replacing notices regularly, which takes time and effort. With the growth of digital technology and the increasing use of smartphones, institutions are gradually adopting electronic systems to make communication faster and more accessible [1].

The use of a digital platform can improve the way information is delivered and managed. By providing notices through an application, users can check updates anytime and from anywhere without depending on physical presence. Digital communication technologies enable information to be transmitted quickly across networks, allowing users to receive updates without delay [2]. Digital systems also allow information to be stored properly so that it can be accessed later when needed. This reduces the use of paper, saves

administrative effort, and ensures that communication is more organized and reliable. Such technology-based solutions are widely encouraged because they improve efficiency and support a more sustainable environment [3].

The Digital Noticeboard System proposed in this project is designed to make communication simple and effective within the institution. Instead of relying on printed notices, administrators can upload announcements digitally, and students can view them through the application. An important feature of the system is the alert message option inside the application, which highlights important notices so that users can easily identify urgent information. This helps students stay informed without repeatedly checking noticeboards or searching through multiple updates. By providing centralized management and organized storage, the system ensures that notices are easy to manage and retrieve when required [4].

However, the existing manual noticeboard system has several limitations. It depends heavily on human effort, requires physical presence to access information, and does not maintain a proper record of previous notices. Students may miss announcements due to lack of timely access, and managing large numbers of printed notices becomes difficult over time. These challenges show the need for a digital solution that can overcome such issues and provide a more convenient way of sharing information [5].

Thus, the proposed Digital Noticeboard System aims to replace the traditional communication method with a simple digital approach that improves accessibility, reduces manual work, and ensures that important information reaches users quickly and effectively [6].

II. LITERATURE SURVEY

Many educational institutions have tried to improve the way they share information by replacing traditional noticeboards with digital solutions. Earlier systems mainly used web-based noticeboards where administrators uploaded announcements to a website, and students had to log in and check updates manually [4]. Although this reduced paperwork, students could still miss important information because they needed to open the website regularly to see whether a new notice had been posted.

Some existing works introduced SMS-based systems to send announcements directly to students' mobile phones [3]. These systems helped deliver information quickly, but they required additional cost for messaging services and were not suitable for sharing detailed notices or documents.

Other approaches used electronic display boards such as LED screens installed within campuses to present notices digitally [1]. These systems allowed information to be updated remotely, but they were still location-based. Students had to be physically present near the display to read the announcements, which limited accessibility.

Mobile application-based notice systems were later developed to allow students to access notices through smartphones [5]. While these applications improved convenience, many of them simply displayed notices

without clearly highlighting which announcements were important. As a result, users had to read through all notices to identify urgent information.

Cloud-based notice management systems were also proposed to provide centralized storage and access across devices [7]. These systems improved data organization, but they often focused more on storage and sharing rather than helping users quickly recognize critical updates.

From the study of these existing systems, it was observed that most solutions concentrated on making notices digital, but they did not provide a simple way to draw attention to important announcements within the application itself. Users were still required to check notices manually, which could lead to missing urgent updates.

Therefore, this project was chosen to design a Digital Noticeboard System that addresses these limitations in a practical and user-friendly manner [8]. In the proposed system, important notices are highlighted through alert messages that can be viewed directly inside the application. This ensures that users can easily identify critical information without searching through all announcements. Along with this feature, the system also provides centralized management, organized storage of notices, and access from any location, thereby overcoming the challenges of manual and location-based communication found in earlier systems.

III. METHODOLOGY

A. TECHNOLOGY STACK

Front-End Technology

The user interface of the Digital Noticeboard System is developed using React Native. This technology allows the application to run on different devices such as mobile phones and computers using a single codebase. It helps in creating a simple and interactive interface so that students can easily view notices and navigate through the application [9]. The design focuses on clarity and ease of use, ensuring that users can quickly understand how to access information without any technical knowledge.

Development Tools

Expo is used as a supporting tool during the development process. It helps in running and testing the application quickly without requiring complex setup. Using Expo, the developer can preview changes instantly, which makes the development process faster and more efficient. These tools provide a smooth environment for building and testing the application.

Back-End Processing

The back-end handles the internal working of the system, such as managing users, storing notices, and controlling access. It ensures that only authorized administrators can add or modify notices, while students

can log in and view them securely. This separation between user interface and system logic improves security and makes the application easier to maintain and update.

Database Design

A database is used to store all information in an organized manner [10]. Instead of keeping records manually, the system saves data digitally so that it can be accessed whenever required. The database contains details such as notice title, description, date, and category. By storing information in a structured format, the system can quickly retrieve or update data without confusion [11].

Two main tables are used in the database. The Users table stores login details and identifies whether the person is an administrator or a student. The Notices table stores all announcements uploaded into the system. This separation avoids duplication and ensures that information is managed properly.

Alert Mechanism for Important Notices

An important feature of this system is the alert message functionality available inside the application. When an administrator marks a notice as important, the system highlights it so that users can easily recognize it. These alerts are displayed within the application itself, allowing students to immediately view critical announcements without searching through all notices. This feature ensures that urgent information is not overlooked.

System Integration

All components of the system work together to provide a smooth experience. The front-end displays information to users, the back-end processes requests, and the database stores and retrieves data. When a new notice is added, it is saved in the database and instantly shown in the application. If it is marked as important, it appears as an alert message for users.

Advantages of the Selected Technology

The chosen technology stack makes the system easy to develop, maintain, and expand in the future. It provides flexibility, reduces manual work, and supports quick access to information. Since the technologies are widely used and reliable, they ensure stable performance and allow future improvements such as adding new features or enhancing security.

Overall, the selected tools and technologies provide a practical and efficient foundation for implementing the Digital Noticeboard System. They enable fast communication, organized data management, and a user-friendly environment for both administrators and students.

B. SYSTEM DESIGN

System Architecture Design

The Digital Noticeboard System is designed to provide an easy way of sharing information between administrators and students. The system follows a simple structure where the user interacts with the application, the application processes the request, and the data is stored in the database. The Administrator and Students access the system through the application interface. The Administrator uploads notices and manages them, while Students log in to view the notices and important alerts.

When a notice is added, the application sends the data to the back-end, which verifies the user and stores the information in the database. When users open the application, the system retrieves the stored data and displays it. Important notices are highlighted as alerts inside the application so that users can easily identify them.

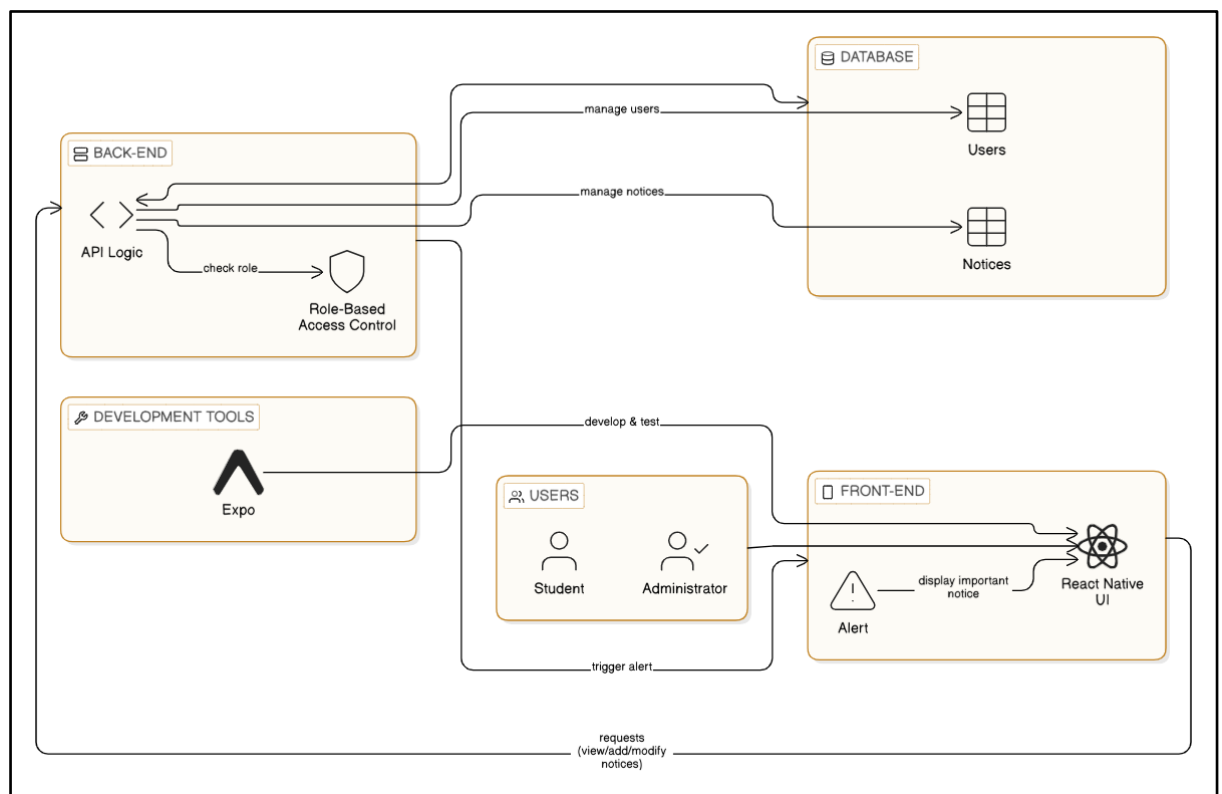


Fig.1: System Architecture of Digital Noticeboard System

Database Design (ER Diagram)

The database is designed to store information in an organized manner so that it can be accessed whenever needed. The system mainly uses two entities: Users and Notices.

The Users entity stores details such as user ID, name, email, password, role, and account creation date. This helps the system identify whether the user is an administrator or a student.

The Notices entity stores notice-related information such as notice ID, title, description, category, importance status, creation date, and the user who created it. Each notice is linked to a user through the “createdBy” field, which shows which administrator uploaded the notice.

There is a one-to-many relationship between Users and Notices, meaning one administrator can create multiple notices, but each notice belongs to only one user. This relationship helps maintain proper data organization.

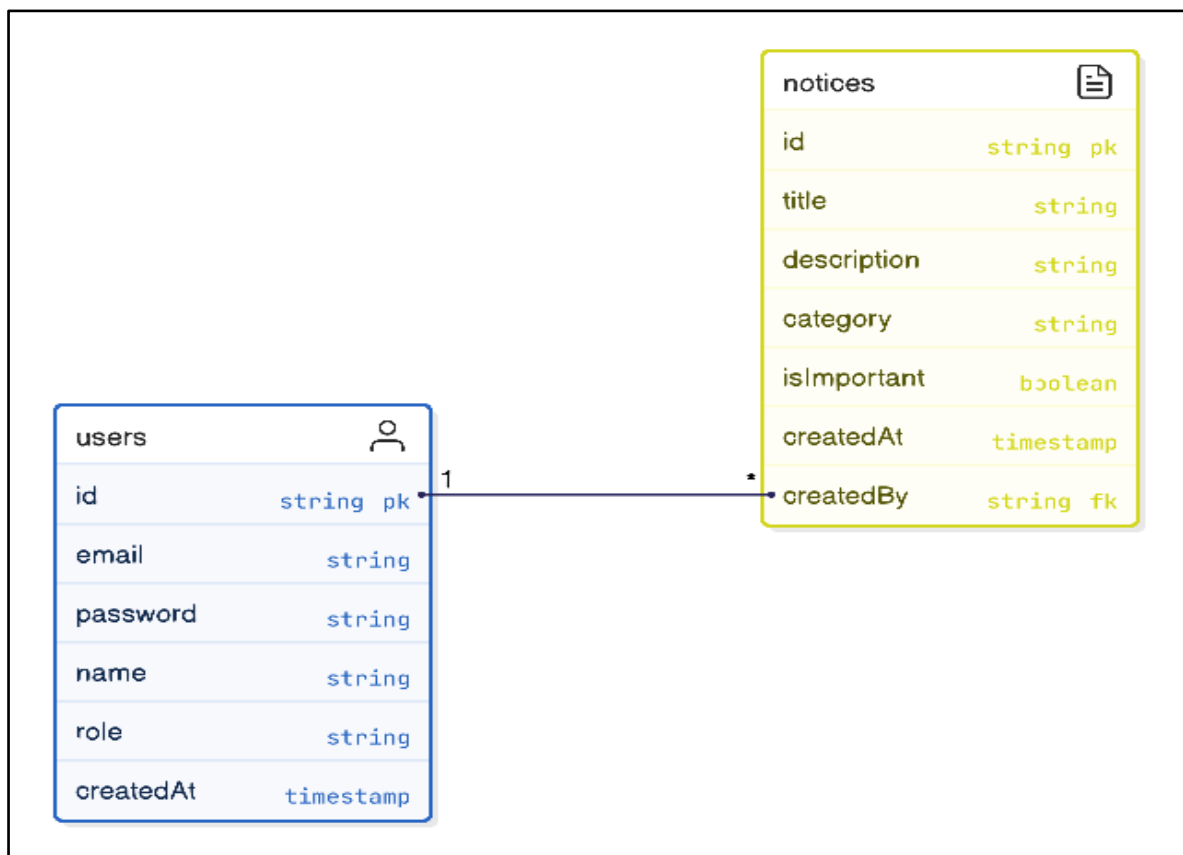


Fig.2: ER Diagram of Digital Noticeboard System

System Modules

The Digital Noticeboard System is divided into different modules to perform specific functions.

The User Authentication Module manages login and verifies user identity. It ensures that only authorized users can access the system.

The Notice Management Module allows administrators to add, edit, and delete notices. It controls how information is created and updated.

The Notice Viewing Module enables students to view notices in a clear format. It retrieves data from the database and displays it on the screen.

The Alert Module highlights important notices so that users can quickly recognize urgent announcements without searching through all notices.

The Database Module stores all information in an organized structure, ensuring that records are maintained properly.

Users of the System

There are two main users in the system: Administrator and Student.

The Administrator is responsible for managing notices. The administrator logs into the system, uploads announcements, marks important notices, and maintains the information.

The Student is the end user who logs into the application to view notices and alerts. Students cannot modify data but can access information anytime.

System Workflow

The system follows a simple workflow. First, the administrator logs into the application and uploads a notice. The system processes the request and stores the notice in the database. If the notice is marked as important, it is identified as an alert.

When students open the application, they can see the list of notices retrieved from the database. Important notices are shown clearly as alerts within the application. This process ensures that communication is fast, organized, and accessible from anywhere.

Overall, the system design focuses on simplicity, organized data management, and easy accessibility. By dividing the system into modules and defining clear roles for users, the Digital Noticeboard System provides an efficient way to manage and share information digitally.

C. IMPLEMENTATION

This chapter explains how the Digital Noticeboard System was developed and what resources were required to run the application. It describes the system requirements, development environment, tools used, and how the different parts of the system were set up and tested.

System Requirements

To develop and run the Digital Noticeboard System, certain hardware and software components are required.

Hardware Requirements:

The system can run on a standard personal computer or laptop. A device with at least 4 GB RAM, a dual-core processor, and basic storage space is sufficient for development and testing. A smartphone or emulator is also required to test the mobile application interface.

Software Requirements:

The application is developed using a modern development environment. The main software includes the operating system (such as Windows 10/11), Node.js for running JavaScript-based applications, and a code editor such as Visual Studio Code for writing and managing the program. These tools help in building and executing the application smoothly.

Development Environment

The system was developed using React Native along with Expo, which allows the application to be tested quickly on mobile devices without complex configuration. Expo provides a simple way to run the application, preview changes instantly, and debug errors during development. This environment makes the development process faster and easier to manage.

Front-End Implementation

The front-end is responsible for the user interface that students and administrators interact with [12]. It includes screens such as login, notice list, and alert display. The interface is designed to be simple so that users can easily navigate through the application. React Native components are used to build responsive layouts, allowing the application to work on different screen sizes.

Back-End Implementation

The back-end manages the internal logic of the system. It processes user authentication, handles notice creation, and ensures that only administrators are allowed to manage notices. Role-based access control is implemented so that students can only view notices, while administrators can add, edit, or delete them. This separation improves security and prevents unauthorized actions.

Database Implementation

The database stores all system data in a structured format. It maintains records of users and notices so that information can be retrieved whenever required. Each notice is stored with attributes such as title, description, category, creation date, and importance status. The relationship between users and notices ensures that each notice is linked to the administrator who created it.

Alert Feature Implementation

A key feature of the system is the alert mechanism for important notices. When an administrator marks a notice as important, the application highlights it as an alert within the system interface. This helps users quickly identify urgent announcements without checking every notice manually. The alert is generated internally by the application and does not rely on external messaging services.

APIs and Libraries Used

The system uses basic APIs and libraries that support application functionality. React Native libraries are used to design the interface and manage navigation between screens. Additional libraries help handle data storage, user input validation, and communication between the front-end and back-end. These libraries reduce development time and ensure stable performance.

Testing and Execution

The application was tested using the Expo environment to ensure that all features work correctly. Functional testing was performed to verify login authentication, notice upload, alert display, and data retrieval. Errors identified during testing were corrected to improve system reliability.

Overall, the implementation process focused on building a simple and functional system that meets the communication needs of the institution. By using suitable tools, structured data storage, and an easy-to-use interface, the Digital Noticeboard System was successfully developed to provide efficient notice management and accessibility.

IV. RESULTS AND DISCUSSION

Administrator Notice Management

The system was tested by logging in as an administrator and uploading notices through the application interface. After entering the required details, the notice was successfully stored in the database and became available for viewing. This confirms that the system is capable of handling data entry, storage, and retrieval without errors. The process was simple and required only a few steps, showing that administrators can manage announcements efficiently.

Create New Notice

Title
URGENT: Exam Schedule Released

Description
Final examination for Semester 1 has been released. All students must check their respective exam dates and venues. Contact the examination cell for any queries.

Category
Urgent

Pin to top ☒

Mark as Important Alert (e.g., Exam) ☒

Alert Expiration Date (Optional)

Cancel Create

Fig. 3: Administrator Uploading Notice

The successful upload demonstrates that the system provides centralized control for managing notices and reduces the need for manual posting.

Viewing Notices in the Application

The application was then tested from the student perspective. After logging in, users were able to view all available notices displayed in an organized format. The system retrieved information directly from the database and presented it clearly on the screen. This shows that the connection between the front-end interface and the database is functioning correctly.

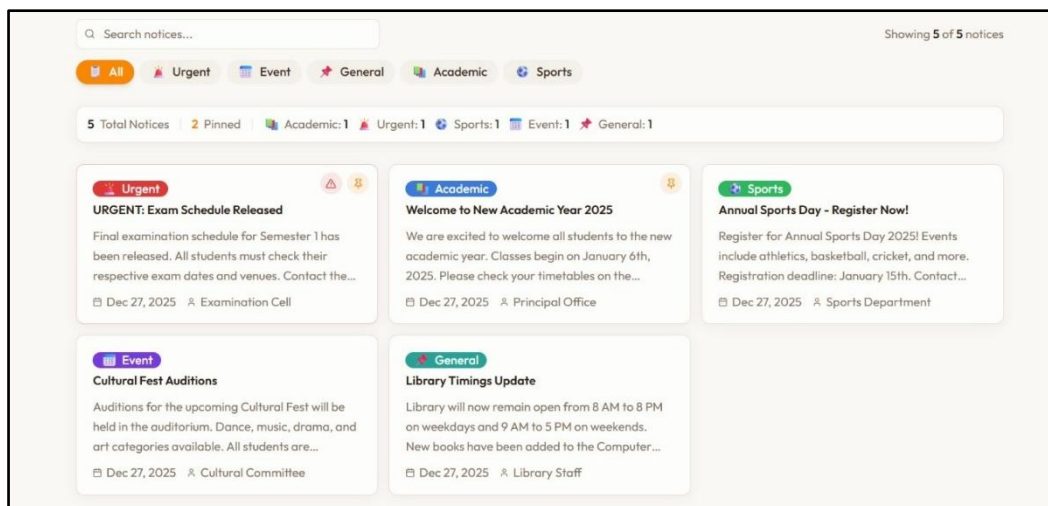


Fig. 4: Notice Display in Application

The display of notices confirms that students can access information easily without visiting a physical noticeboard.

Alert Display for Important Notices

To test the alert feature, a notice was marked as important by the administrator. The system highlighted this notice inside the application so that it could be easily recognized by users. The alert was shown immediately when the application was opened, ensuring that critical information receives proper attention.

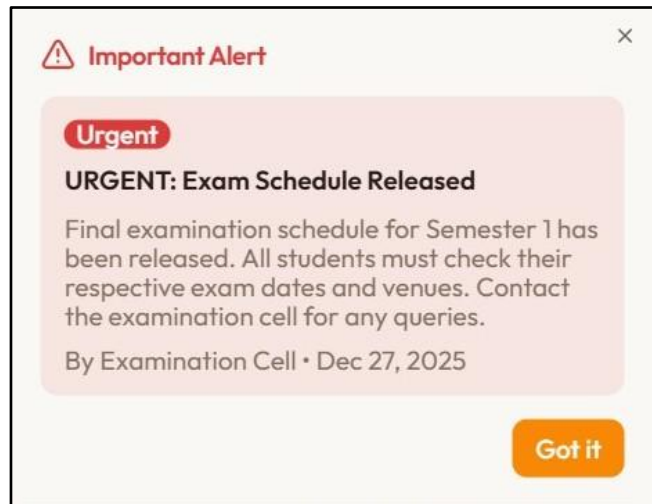


Fig. 5: Important Notice Alert

This result shows that the alert mechanism helps users identify urgent announcements quickly and prevents important information from being overlooked.

Discussion of System Performance

The testing of all modules shows that the Digital Noticeboard System operates as intended. The system allows administrators to manage notices digitally while enabling students to access information anytime. The use of structured storage ensures that notices are maintained properly and can be retrieved whenever required. The application reduces manual effort, minimizes paper usage, and improves communication within the institution. Overall, the system provides a reliable and user-friendly solution that successfully replaces the limitations of the traditional noticeboard.

V. CONCLUSION

The Digital Noticeboard System was developed to provide a simple and effective solution for sharing information within an educational institution. Traditional noticeboards require manual effort and physical presence to access information, which often leads to missed announcements and difficulty in managing notices. This project aimed to overcome these challenges by introducing a digital platform that allows notices to be uploaded, stored, and viewed through an application.

The system successfully enables administrators to manage announcements in a centralized manner while allowing students to access information anytime and from any location. The integration of a structured

database ensures that notices are stored in an organized way and can be retrieved when needed. The alert feature included in the application helps highlight important notices so that users can easily identify urgent updates.

The implementation of this system reduces the need for paper-based communication and minimizes manual work, making the process faster and more efficient. The user-friendly interface ensures that the application can be used without technical difficulty, allowing both administrators and students to interact with the system comfortably.

Overall, the Digital Noticeboard System demonstrates how digital technology can replace traditional communication methods and improve the flow of information within an institution. The project achieves its objective of providing a reliable, accessible, and organized platform for notice management, thereby enhancing communication and supporting a more modern and efficient academic environment.

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